# The AI-GPR Index: Measuring Geopolitical Risk using Artificial Intelligence*

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**Abstract**

We use a large language model to read nearly 5 million newspaper articles since 1960 and measure geopolitical risks in ways that keyword counts cannot: who threatens whom, whether a danger has already materialized, and whose oil supply is at risk. Applying GPT-4o mini to the *New York Times*, *Washington Post*, and *Chicago Tribune*, we build a daily index of geopolitical risk together with measures of threats and acts, of oil supply disruptions by region, and of risk directed from one country to another. Against human coders, the model misses far fewer relevant articles than the keyword approach and flags fewer irrelevant ones. The new measures deliver results that keyword indexes cannot: stock markets price threats and acts differently, geopolitical oil market shocks can be separated from other oil price movements, and a rise in geopolitical risk between two countries is followed by a decline in their trade that builds over two years.

**JEL Classification:** D80, E66, F51, G15.

**Keywords:** Geopolitical Risk, Textual Analysis, Large Language Models, Measurement Error, Content Classification.

## 1 Introduction

Geopolitical risk—wars, terrorism, and tensions between nations—is an important driver of macroeconomic fluctuations. Caldara and Iacoviello (2022) measured it with the Geopolitical Risk (GPR) Index, which counts newspaper articles containing combinations of geopolitical and risk-related

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keywords.[1] The index has become a standard benchmark for the risks that shape trade patterns, investor decisions, and asset prices. Keyword counts, however, can only tell how much the news is about war and terrorism. They struggle with articles that mention a conflict in passing or in a historical context, and they say little about what an article describes.

Many of the questions economists ask about geopolitical risk depend on exactly that. Is the danger a threat, or an event that has already happened? Whose oil supply is at risk? Which country is acting against which? A keyword search finds articles that mention both the United States and Iran, but not who is threatening whom. Answering such questions requires reading the articles, which large language models (LLMs) now make possible at the scale of a newspaper archive.

We use an LLM, OpenAI's GPT-4o mini, to read nearly 5 million articles from the *New York Times*, *Washington Post*, and *Chicago Tribune* from 1960 to the present. We start with an index of overall geopolitical risk, the AI-GPR. It keeps the definition of geopolitical risk behind the original GPR but scores each article on a continuous scale, so that the index reflects the severity of events as the news presents them. Checked against human coders, the LLM misses far fewer relevant articles than the keyword approach and flags fewer irrelevant ones. The daily index also moves around much less from one day to the next. We then ask further questions of the same articles, each requiring only a new prompt rather than a new dictionary. We split geopolitical risk into threats and acts, identify articles about geopolitical disruptions to oil supply and the regions affected, and record which countries are involved in each event and whether a country is the initiator, the respondent, or a spillover party. The last step yields country indexes for two hundred nations and directed indexes for pairs of countries, which keyword methods cannot produce.

These measures afford results that keyword indexes cannot deliver. First, we revisit the link between geopolitical risk and financial markets. The AI-GPR gives a more precisely estimated negative effect of geopolitical risk on stock returns than the keyword index, particularly for its persistent component. Splitting the index into threats and acts shows that markets price the gradual and persistent buildup of threats but respond to acts only insofar as they are unanticipated.

Second, the Oil GPR index, built from articles about geopolitical oil supply disruptions, helps separate geopolitical oil market shocks from other oil price movements. We identify these shocks with a proxy structural VAR in the spirit of Verduzco-Bustos and Zanetti (2026), replacing their GPR instrument with oil price surprises on days when threats of geopolitical oil supply disruptions spike. The identified shock generates a persistent rise in oil prices, a rise in inflation and long-term yields, and a decline in economic activity.

Third, we use the bilateral indexes to study the response of international trade to geopolitical risks. A rise in geopolitical risk between two countries is followed by a decline in their trade that

[1] The use of dictionary-based methods itself builds on the foundational work of Baker, Bloom, and Davis (2016) and Hassan, Hollander, van Lent, and Tahoun (2019), among others.

builds over two years and then fades.

Several of the measures introduced in this paper are ones we have wanted to build since the original GPR index appeared. Researchers, policymakers, and market participants who used it kept asking for series for a large number of countries, for risks to oil supply, and for who is acting against whom, and keyword counts could deliver some of them only roughly and others not at all. The country-level, oil-supply, and bilateral series in this paper answer those requests. We make all of them available on a companion webpage, where they are updated monthly.

Our work contributes to a rapidly emerging literature using LLMs for economic measurement (e.g., Ito, Sato, and Ota, 2025 and Clayton, Coppola, Maggiori, and Schreger, 2025). Keyword searches remain useful as a cheap first filter, as in our own pipeline. Our focus on geopolitical risk also connects our work to the growing literature on geoeconomics (see Mohr and Trebesch, 2025 and Clayton, Maggiori, and Schreger, 2026) and provides an improved set of data to make sense of geopolitical risk both within and across country borders.

Section 2 describes the data and the LLM-based methodology. Section 3 presents the AI-GPR index. Section 4 assesses the index's validity and robustness, contrasting it with keyword-based classification, validating it against human coding, and confirming robustness to the choice of specific LLM used for classification. Section 5 introduces three further dimensions of measurement: threats and acts, oil, and country and bilateral risk. Section 6 presents our empirical applications, and Section 7 concludes.

## 2 Data and Methodology

### 2.1 Data Sources

Our construction of the AI-GPR index samples newspaper data from the *New York Times*, *Washington Post*, and *Chicago Tribune* from 1960 to the present. We restrict our attention to articles categorized as news content rather than advertisements or non-editorial material. We maintain the AI-GPR as a live index, so new articles are scored with the same query, prompt, and model, and posted to the webpage each month. For descriptive statistics and validation checks presented in the paper, we restrict our sample to articles published between 1960 and 2025.

To construct the index, we adopt a two-stage process that first filters for articles likely to discuss geopolitical events and then passes them through an LLM for scoring. The keyword screen filters out articles with no geopolitical content, and the LLM then scores the remaining articles on a scale from 0 to 1. We use the following query on the ProQuest TDM Studio database as our first-stage filter for articles likely to discuss geopolitical risk:

```
SEARCH QUERY:

PUBDATE(>=19600101) AND PUBDATE(<=20251231) AND LA(English) AND (airstrike*
OR alliance OR annex* OR attack* OR blockade* OR bomb* OR cease-fire OR
combat OR confrontation OR conflict* OR coup OR crisis OR diplomat* OR
diplomacy OR embargo OR enem* OR hostage* OR hostil* OR instability OR
invade* OR invasion* OR militia OR military OR missile* OR nuclear OR
refugee* OR riot* OR sanction* OR sovereign* OR terror* OR treaty OR troop*
OR truce OR unrest OR violence OR war OR weapon*) AND (publication("New
York Times") OR publication("Washington Post") OR publication("Chicago
Tribune")) AND DTYPE(article OR commentary OR editorial OR feature OR
'front page article' OR 'front page/cover story' OR news OR report OR
review)
```

The query looks for articles containing at least one keyword potentially related to geopolitical risk. These keywords span three categories of geopolitical risk: direct conflict and military action (war, invasion*, attack*, terror*, conflict*, ...), political crisis and instability (crisis, instability, unrest, coup...), and strategic responses and diplomacy (sanction*, embargo, blockade*...). Our search returns 4.6 million articles from the universe of articles published across the three publications over the sample period. Of the 4.6 million articles, 1.2 million (about a quarter) receive a positive GPR score from the LLM. Our search retrieves 54 percent of all the articles published in the three newspapers over the sample period.[2] Section 4.2 verifies that the keyword filter has a negligible false-omission rate. For each selected article, we extract metadata (publication date, headline) and its textual content from the ProQuest database. To manage computational costs, we truncate article text to the first 2,000 characters.[3]

## 2.2 LLM Classification: Definition, Prompt, Implementation

We use Caldara and Iacoviello (2022)'s definition of geopolitical risk, which encompasses wars, escalation of existing events, terrorist attacks, and diplomatic tensions. It covers current risks and events, not historical retrospectives, fictional depictions, or metaphorical uses of geopolitical terminology. We use the following system prompt to instruct the LLM to evaluate articles:

[2] We designed our query around words that align with the definition of geopolitical risk, even if the query does not strictly maximize recall per article retrieved.

[3] As we show in Section 4.2, scores based on truncated text are nearly identical to those based on the full article.

```
CLASSIFICATION PROMPT:
You will be given a news article.  Classify the article's assessment of
geopolitical risk based only on what the article states or strongly implies.

Geopolitical risk is defined as the threat, realization, and escalation of adverse
events associated with Wars (initiation of new conflicts); Escalation of existing
wars; Major Terrorist attacks; Tensions between nations or involving cross-border
political actors that affect international relations.

Assign a geopolitical risk score from 0.0 to 1.0 based on the following scale:
0.0--0.2:  No mention of geopolitical risks.
0.2--0.4:  Mentions of minor tensions, diplomatic disputes, or isolated incidents
with limited escalation risk.
0.4--0.6:  Discussion of significant tensions, ongoing regional conflicts, or
moderate escalation risk; some concern about war or terrorism.
0.6--0.8:  Substantial discussion of major war initiation/escalation risks, active
terrorism threats, or high likelihood of significant escalation.
0.8--1.0:  Extensive coverage of imminent or new war, major war escalation, severe
terrorism threats, or critical threat to international stability.

Movies or books about geopolitical risks, stories remembering old events,
obituaries, anniversaries of old events should be given 0 unless they explicitly
mention current risks.
Note:  Purely domestic political events (elections, protests, internal policy
debates) should score 0.0 unless they have implications for international tensions
or cross-border conflicts.
```

We use zero-shot prompting where the model receives explicit criteria and a scoring scale but no labeled examples.[4] The scale from 0.0 to 1.0 allows the model to rate articles with varying intensities of geopolitical risk rather than binary classification. We set the LLM's temperature parameter to zero, which minimizes stochastic variation in responses across runs. We use structured output to return model output in JSON format, making extraction and processing of scores more efficient. For each article, we construct a request consisting of the system prompt, the article's title, and its truncated text. All scores fall within the specified 0.0–1.0 range.

### 2.3 Index Construction

We construct the AI-GPR index by aggregating individual article scores at daily frequency. For each date $t$, we compute:

$$\text{AI-GPR}_t = \frac{1}{\bar{S}} \times \frac{1}{A_t} \sum_{i=1}^{N_t} S_{it} \tag{1}$$

where $S_{it}$ represents the geopolitical risk score for article $i$ at time $t$, $N_t$ is the total number of

[4] One-shot or few-shot prompting would require embedding one or more example articles in every request; at the scale of approximately 4.6 million articles, even a single example would roughly double input token costs.

scored articles published on date $t$, $A_t$ is the total number of articles published on date $t$ across all three newspapers (not just those matching geopolitical keywords), and $\bar{S}$ is a normalization constant chosen such that the index has a mean of 100 over the 1985–2019 period. Dividing by $A_t$ accounts for variation in newspaper output over time and follows the original GPR index.[5]

The index measures the intensity-weighted share of newspaper coverage devoted to geopolitical risk. The numerator sums the risk intensity of each article while the denominator scales by total newspaper output. In Appendix A.2 we show that the index is robust to the main construction choices, particularly whether article scores enter on the intensive or the extensive margin. Because the index is daily, it records short-lived spikes and dates events precisely, which matters for event studies and for measuring the immediate market reaction to geopolitical developments.

### 2.4 Computational Considerations

A practical concern for LLM-based approaches is computational cost. At current pricing, processing one article through GPT-4o mini costs approximately \$0.0001 for our truncated text length. For our sample of approximately 4.6 million articles from 1960–2025, the total computational cost is approximately \$450. Processing time is also manageable. Scoring one article takes roughly one second. With five API calls running in parallel, scoring the full corpus of 4.6 million articles takes about 250 hours.

## 3 The AI-GPR Index

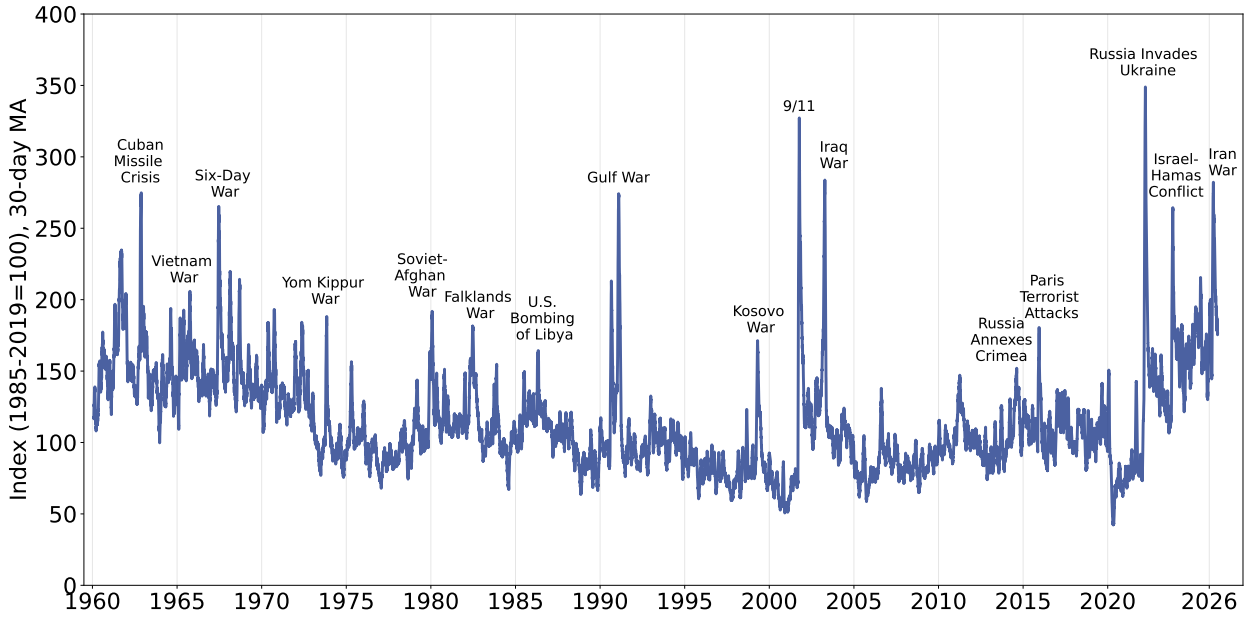
Figure 1 displays the AI-GPR index over the full sample, with labels marking the major geopolitical episodes of the past six decades. The index rises during every major war and crisis, and its highest readings come around the Gulf War, the September 11 attacks, Russia's invasion of Ukraine, and the 2026 U.S. military intervention in Iran.

Table 1 presents summary statistics for the AI-GPR index alongside the keyword-based GPR index over the 1960–2025 period as a benchmark. The keyword-based index applies the Caldara and Iacoviello (2022) methodology—which classifies articles using proximity searches of geopolitical and risk-related terms—to the same three newspapers used for the AI-GPR (*New York Times*, *Washington Post*, and *Chicago Tribune*).[6] The AI-GPR index is smoother than the original GPR, with a lower standard deviation and a thinner right tail. The first percentile is well above zero,

[5] The denominator $A_t$ counts articles containing seven common words ("the," "be," "to," "of," "and," "at," and "in"), which excludes non-editorial content such as captions, advertisements, and stock quotes. It totals about 8.5 million articles over the sample.

[6] The original GPR comes in two versions: a historical index based on three newspapers that extends back to 1900, and a recent index based on ten newspapers that begins in 1985. Our version is more comparable with the historical version as we sample from the same three newspapers, but our index begins in 1960. The full search string is reported in Appendix A.5.

Figure 1: The AI-GPR Index




*Notes*: The AI-GPR index plotted as a 30-day moving average. The index sums LLM-assigned geopolitical risk scores (0–1) across all articles each day, normalized by total newspaper article count, and is scaled to a mean of 100 over 1985–2019. Regularly updated data are available on the companion webpage. Source: authors' calculations. Last observation: June 2026.

with no single day recording zero geopolitical content. The AI-GPR is also more persistent, with a higher 90-day autocorrelation.

On an average day, the AI-GPR classifies 15 percent of all articles as discussing geopolitical risk, against 3.3 percent under the keyword-based approach. This fits the finding of Carlson and Dell (2025) that keyword classification understates geopolitical risk: when they correct the keyword-based GPR with a human-labeled sample, the corrected series is higher than the original. The two indexes nevertheless remain substantially correlated, suggesting that while the keyword approach may understate the share of geopolitically relevant newspaper articles, it still captures the same peaks and troughs.

Figure 2 plots the AI-GPR alongside the original GPR index at monthly frequency. The two series co-move closely but diverge in several periods. In the 1960s–1970s, the AI-GPR index is elevated relative to the keyword-based GPR, perhaps because the LLM identifies geopolitical tension in words and phrases outside the keyword dictionary. The Cold War-era tensions, the Vietnam War, and the Six-Day and Yom Kippur wars all register prominently. In the 1990s, the two indexes move together during the Bosnian War, but the AI-GPR rises well above the keyword index during the Kosovo War, suggesting that the LLM captures the geopolitical significance of NATO's intervention more fully than keyword counting. In the 2000s, both indexes spike sharply after the September 11

Table 1: AI-GPR Index: Summary Statistics

| **Statistic** | **AI-GPR** | **GPR (Original)** |
|---|---|---|
| Total newspaper articles | 8,514,822 | 8,514,822 |
| Articles sampled | 4,571,496 | — |
| Articles scored positive ($> 0$) | 1,166,083 | 262,096 |
| Standard Deviation | 48.52 | 60.55 |
| Skewness | 1.64 | 1.97 |
| 1st Percentile | 41.97 | 18.14 |
| Median | 105.46 | 99.15 |
| 99th Percentile | 266.95 | 300.16 |
| 90-day Autocorrelation | 0.73 | 0.62 |
| Correlation with Original GPR | 0.69 | 1.00 |
| Positive Articles Share (%) | 15.00 | 3.28 |
| Positive Among Sampled (%) | 26.16 | 3.28 |

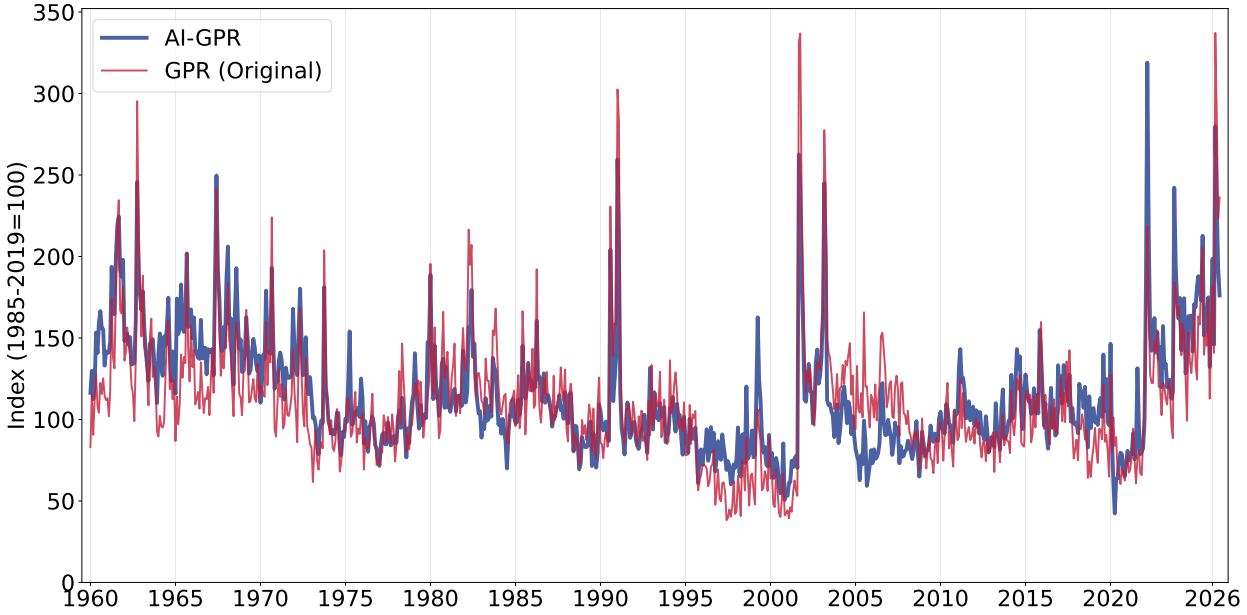
*Notes*: Summary statistics computed over the 1960–2025 sample. The "GPR (Original)" column is the historical GPR index of Caldara and Iacoviello (2022), computed on the same three newspapers as the AI-GPR over 1960–2025. Both indexes are normalized to a mean of 100 over 1985–2019. The 90-day autocorrelation is the correlation between the 90-day moving average on date $t$ and on date $t-90$. Positive Articles Share (%) is the average daily share of all newspaper articles classified as containing geopolitical risk: for the AI-GPR, articles receiving a positive score; for the original GPR, articles matching the keyword proximity search. Positive Among Sampled (%) is the average daily share of positive articles among those in the queried sample: for the AI-GPR, the share of articles sent to the LLM that receive a score $> 0$; for the original GPR, this coincides with Positive Articles Share (%) since there is no separate sampling stage.

attacks and during the Iraq War. The AI-GPR declines somewhat faster after the initial Iraq War spike, consistent with the LLM distinguishing articles that cover active conflict from articles that mention the Iraq War in other contexts. From 2010 onward, the indexes track each other closely, though the AI-GPR is somewhat more elevated in recent years.

The differences between the two indexes are more visible at daily frequency around key geopolitical events, as illustrated in Figure 3.[7] Both indexes spike at the onset of each event, but in every episode the AI-GPR is smoother from one day to the next. A likely reason is that the LLM recognizes geopolitical risk independent of how journalists describe it, whereas the keyword index is more sensitive to writing style and counts an article only when specific terms appear. Lower noise matters most for high-frequency applications, such as the weekly stock-return regressions in Section 6.1, where noise in the index would weaken the estimated effect of geopolitical risk.

[7] Both indexes are adjusted for day-of-the-week effects. Newspapers publish more articles on some days of the week than on others, which creates a weekly pattern in both daily indexes. We remove it by dividing each index by a day-of-the-week factor estimated over the previous year.

Figure 2: AI-GPR and Original GPR Index




*Notes*: The AI-GPR index (blue, thick) and the original GPR index (red, thin), both plotted at monthly frequency. The original GPR here is the three-newspaper version of the Caldara and Iacoviello (2022) index—their historical index, which extends back to 1900—computed on the same three newspapers as the AI-GPR; the comparison is with this version rather than their ten-newspaper index, which begins in 1985. Both indexes are normalized to a mean of 100 over 1985–2019. The AI-GPR index sums LLM-assigned geopolitical risk scores across all articles each day, normalized by total newspaper article count, then averaged to monthly frequency. The original GPR counts articles matching specific keyword proximity searches. Source: authors' calculations. Last observation: June 2026.

# 4 Validation and Robustness

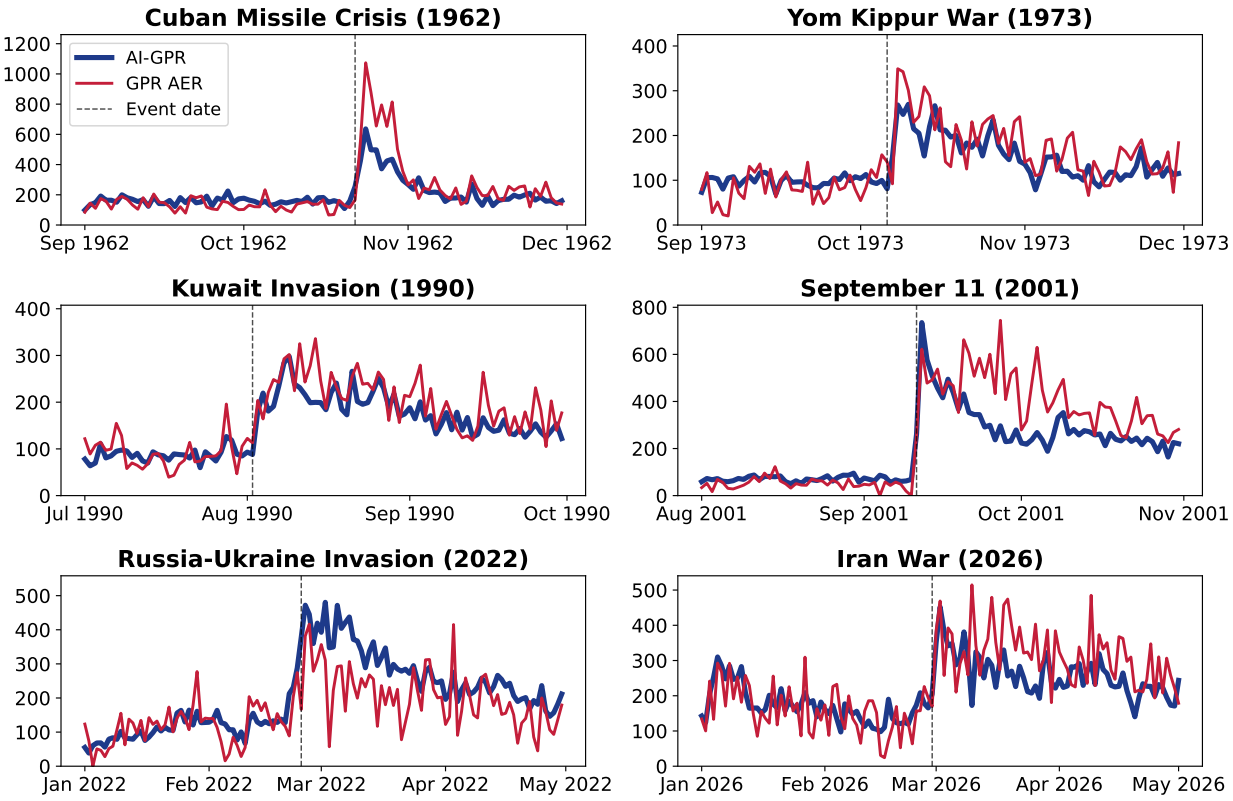
This section compares the LLM classifier with keyword-based selection and validates both against human coding, quantifies the remaining sources of measurement error, and checks that the index is insensitive to the choice of language model.

## 4.1 Keywords, the LLM, and Human Judgment

The keyword-based GPR flags an article when it contains specific word combinations, for example "war" near "threat" or near "began."[8] This approach has two main weaknesses. First, the classification is binary: every matching article counts once, whether geopolitical risk is its main subject or a passing mention. Second, the query depends on its vocabulary: an article about a "military standoff" is missed if this word combination is not in the dictionary. The keyword-based query also allows for exclusion words that remove some false positives, such as obituaries or film reviews, but cannot catch them all. By contrast, the LLM reads the whole passage. It recognizes geopolitical

[8] The full query is reproduced in Appendix A.5.

Figure 3: AI-GPR vs. Original GPR Around Key Geopolitical Events




*Notes*: Each panel shows the daily AI-GPR index (blue, thick) and original GPR index (red, thin) around a major geopolitical event. Vertical dashed lines mark the event date. Both indexes are at daily frequency and normalized to a mean of 100 over 1985–2019. Both daily indexes are adjusted for day-of-the-week effects using trailing one-year averages of multiplicative day-of-the-week dummies. Source: authors' calculations.

risk however it is described, grades each article by how central and severe that risk is, and can better tell a war movie from an actual war.

The cost of an LLM is transparency. Unlike a keyword list, an LLM cannot be audited by reading its rules. Explicit criteria in the prompt and a temperature of zero mitigate this problem, but the real check is to compare its output against the judgment of human readers. To this end, we hand-score a sample of 2,447 articles from all three newspapers, using the same 0-to-1 scale given to the LLM.[9] The sample is stratified so that it includes articles the keyword filter never retrieves as well as those it does, and we weight results by each stratum's share of the corpus, so they describe the whole newspaper corpus rather than only the articles that the model scored.

Our main finding is that the LLM scores closely match human judgment. Most articles are read and scored by two human coders, which lets us compare the LLM with each coder and the coders

[9] Full details of the audit, including the sampling design, the cross-tabulation of human and machine scores, the comparison with the keyword query, and illustrative cases, are reported in Appendix A.3.

with each other. The model's scores have a correlation of 0.82 with the individual human scores. On the same articles, the two coders' scores have a correlation of 0.78 with each other. The model therefore agrees with a human coder more closely than two human coders agree with each other. The model's and the humans' average scores are also nearly identical, so the classifier introduces no aggregate bias in levels.

Occasionally, the model and the humans disagree, although the differences in the scores are small. Disagreements fall into a few patterns. The model sometimes treats cross-border economic disputes, such as tariffs or currency crises, and purely domestic unrest, such as riots, as geopolitical risk, although the rubric reserves the concept for war, terrorism, and military tension. Additionally, the LLM sometimes scores recent retrospective coverage as if the events were current, although it occasionally underscores the immediate aftermath of a war outbreak or terrorist attack, so the two errors partly offset. Finally, some disagreements arise when a human reads risk into an article that only implies it; these reflect ambiguity in the article and the rubric rather than model error.

The LLM beats the original keyword query on both counts: it flags fewer irrelevant articles, and it misses far fewer relevant ones. Of the articles the LLM flags as geopolitical, human coders consider 13 percent irrelevant, against 18 percent for the keyword query.[10] Of the articles human coders consider relevant, the LLM misses about one in four, while the keyword query misses more than four in five. This pattern mirrors what Hartley (2025) documents for the Economic Policy Uncertainty index of Baker, Bloom, and Davis (2016), where language-model classifiers raise recall from 45 to 82 percent and also improve precision. The advantage of reading over keyword matching thus appears to be a general feature of news-based indexes.

### 4.2 Measurement Error

We test four potential sources of measurement error in the AI-GPR index: model stochasticity, text truncation, attenuation from the two-stage filtering process, and hindsight knowledge embedded in the language model.

**Model stochasticity.** Even at temperature zero, LLMs are not deterministic and may produce slightly different outputs across calls. To quantify this, we run GPT-4o mini three times on the same subset of 1,050 articles. Scores are nearly identical across the three runs and the vast majority of articles receive exactly the same score. Raising the temperature to 0.5 and 1.0 also seems to have a negligible impact as within- and cross-temperature agreement remains high and mean scores are nearly the same.[11]

[10] Both rates are computed over individual article–coder ratings rather than the coders' consensus, so an article that one coder scores as zero and the other as positive is not averaged into a positive benchmark. This matches the definition behind the false-positive rate reported for the original GPR.

[11] Across the three temperature-zero runs, the pairwise correlation is 0.99.

**Text length.** Our baseline classifier truncates article text at 2,000 characters to reduce processing time and cost. To verify that this truncation does not materially affect classification, we re-score the same articles using their complete text. Truncated and full-text scores are highly correlated and identical for most articles. Full-text scores are slightly higher on average, as longer articles carry additional geopolitical content beyond the opening paragraphs. The ranking of articles is nearly unchanged.[12]

**First-stage attenuation.** The keyword pre-filter could introduce measurement error by missing geopolitically relevant articles or by including irrelevant ones. We assess the false-omission rate—the share of the articles the filter discards that are in fact geopolitically relevant—by applying the LLM classifier to a random sample of ten thousand articles across all three newspapers that did not match any of our search keywords. Only 0.6 percent received a positive geopolitical risk score, and the few that did scored low. No article scored higher than 0.4 on the scale, and the discarded articles taken together would add almost nothing to the index.[13]

**Hindsight knowledge.** Because GPT-4o mini is trained on text through October 2023 (OpenAI, 2024), it scores historical articles already knowing how the events they describe unfolded. The model could implicitly exploit this knowledge and misstate the risk perceived by a contemporaneous reader. To test this, we re-score the articles surrounding four major events—the Cuban Missile Crisis, the 1990 invasion of Kuwait, the September 11 attacks, and Russia's 2022 invasion of Ukraine—with a modified prompt that fixes each article's own publication date as the present ("Today is {date}"), supplies the date and headline, and explicitly instructs the model that it has no knowledge of events occurring after that date.[14] We find that, across all episodes, the "hindsight-robust" index is nearly indistinguishable from the baseline. We also re-score the roughly 2,000 human-audit articles, which span the full sample period, with the date-anchored prompt; their scores barely change.[15]

### 4.3 Robustness to AI Model Choice

To assess whether our choice of LLM affects the index, we score a sample of 2,050 articles with four OpenAI models (GPT-4o mini, our baseline; GPT-4o; GPT-5-mini; and GPT-5) and one open-

[12] The correlation between truncated and full-text scores is 0.93, with 86.7 percent of articles scored identically; the mean score rises from 0.129 to 0.153 with full text.

[13] Because this check relies on the model's own judgment, we also read 400 of the discarded articles without seeing the model's scores. We found that fewer than two percent of articles contained any geopolitical risk content; details are in Appendix A.3.

[14] An alternative would be to follow the lead of He, Lv, Manela, and Wu (2026), who train language models only on text available up to each date, which rules out hindsight by construction.

[15] Appendix A.4 reproduces the full prompt and plots the comparison.

weight model, Llama 3.1 8B Instruct. We set the temperature to zero for the models that allow it.[16] The scores of GPT-4o mini are highly correlated with those of every alternative.[17] Since it gives similar scores at a fraction of the cost of the larger models, we use GPT-4o mini as our baseline. The agreement with Llama also matters for reproducibility. Proprietary models are periodically retired, a concern Carlson and Dell (2025) raise for AI-derived measures, but Llama's weights are permanently and publicly available, so the index can be closely reconstructed with it.[18]

# 5 The Anatomy of Geopolitical Risk

We apply further LLM classifiers to measure three additional dimensions of geopolitical risk: threats and acts, oil, and country-level and bilateral risk. Each is produced by applying an additional classification layer to the articles that the baseline classifier identifies as containing geopolitical risk. To be clear, some of these splits could be done with a careful choice of keywords: the original GPR already separates threats from acts using different groups of search terms, and a list of oil terms finds most articles about oil. However, other questions depend more subtly on how the parts of an article relate to each other. A keyword search can find an article that mentions Russia, Ukraine, and oil, but it cannot tell which country attacked which, or whether the risk is to Russian oil exports or to supplies from the Middle East. Answering such questions with dictionaries would require a separate set of rules for every region and every country pair. With an LLM, each new question takes one more prompt, as we show next.[19]

## 5.1 Threats and Acts

A key innovation of Caldara and Iacoviello (2022) was to separate geopolitical risk into threats—adverse events that are anticipated, feared, or warned of but have not yet materialized—and acts—events that have actually occurred or are ongoing. Our analysis of stock returns in Section 6.1 shows that markets price the two differently.

We build an analogous decomposition of the AI-GPR index with a second LLM classification layer that reads each article and judges the nature of the risk described. We ask GPT-4o mini to

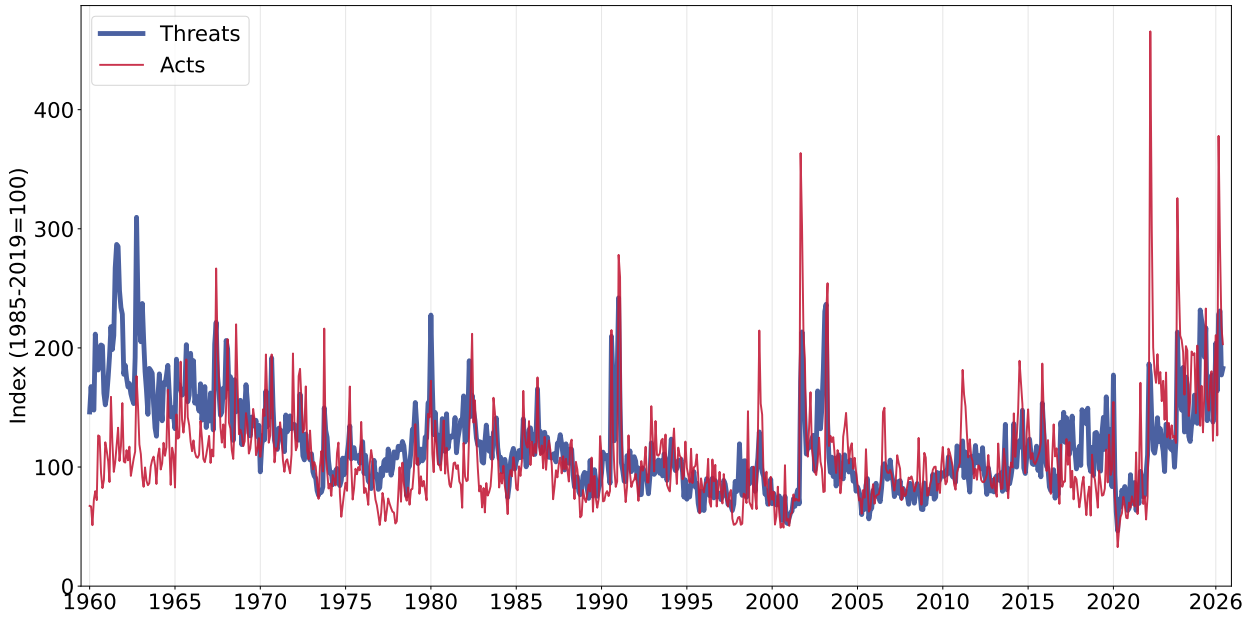
[16] GPT-5-mini and GPT-5 do not allow the temperature to be set, so they run at their default setting with reasoning effort set to high. For Llama, we download the model weights (meta-llama/Llama-3.1-8B-Instruct) from Hugging Face and run them on an Amazon SageMaker cloud server, with 8-bit quantization and greedy decoding, which is equivalent to a temperature of zero.

[17] The pairwise correlations between GPT-4o mini and the alternatives are 0.87 (GPT-4o), 0.87 (GPT-5-mini), 0.87 (GPT-5), and 0.91 (Llama). The positive-classification share ranges from 25 to 33 percent across models.

[18] We also tried to improve the open-weight model by training it further on up to 300 articles scored by our human coders. On a separate set of articles, its scores agreed less with human judgment, not more: the correlation fell from 0.83 to between 0.45 and 0.70, and in most runs the retrained model flagged far fewer articles as containing geopolitical risk.

[19] Every subindex in this section divides the relevant scores by $A_t$, the total article count in equation (1), so all subindexes share the scale of the aggregate AI-GPR. The classification prompts are reproduced in Appendix A.7.

Figure 4: Geopolitical Threats and Acts




*Notes*: The threats subindex (blue, thick) and acts subindex (red, thin) of the AI-GPR index, plotted at monthly frequency. A second LLM classification layer assigns each geopolitically relevant article (AI-GPR score $> 0$) to "threat," "act," or "both"; articles labeled "both" contribute to each series. Series are indexed so that each series averages 100 over 1985–2019. Source: authors' calculations. Last observation: June 2026.

read again every article with a positive AI-GPR score.[20] It places each article into one of three categories. An article is a threat if it mostly centers on risks, warnings, or tensions that have not materialized yet, and an act if it reports an adverse event that has occurred, such as a terrorist attack, the start of a war, or a coup. A third category covers articles that discuss a realized event together with the risk of further escalation. We count these toward both subindexes, so the threat series adds them to the threats and the act series adds them to the acts. Each subindex is then built exactly like the headline AI-GPR index: we sum the geopolitical risk scores of its articles, divide by total newspaper volume, and rescale the series to average 100 over 1985–2019.

Figure 4 plots the two subindexes. Acts spike sharply around the realization of major adverse events, such as the 1991 Gulf War, the September 11 attacks, the 2003 Iraq War, and Russia's 2022 invasion of Ukraine. Threats are higher during the Cold War in the 1960s and other spells of elevated tensions, when coverage is dominated by warnings and brinkmanship. Threats also run ahead of acts in the late 2010s, when coverage centered on North Korea's nuclear program and the Iran nuclear deal, neither of which turned into open conflict. Since 2020 the balance has tipped back toward acts, as the wars in Ukraine, Gaza, and Iran have come to dominate the news.

[20] As in the baseline scoring step, the model reads the first 2,000 characters of each article.

### 5.2 Oil Markets

We construct a second subindex identifying oil market disruptions driven by geopolitical events using a two-step classification process. In the first step, to economize on classification costs, we restrict attention to articles with a GPR score above 0.5 that mention at least one oil- or energy-related keyword. Articles without such keywords are assumed not to mention oil-disruption risk. In the second step, we apply a second LLM prompt to the retained articles.[21] The prompt asks GPT-4o mini to determine whether the article discusses an oil or energy disruption caused by a geopolitical event and, if so, which geographic region(s) are affected from a predefined list: Middle East, Russia, USA, Venezuela, North Africa, West Africa, Central Asia, North Sea, Canada, Mexico, Latin America, Southeast Asia, and China.

For each date $t$, we construct the Oil GPR index as follows:

$$\text{Oil GPR}_t = \frac{\sum_{i \in D_t} S_{it}}{A_t} \tag{2}$$

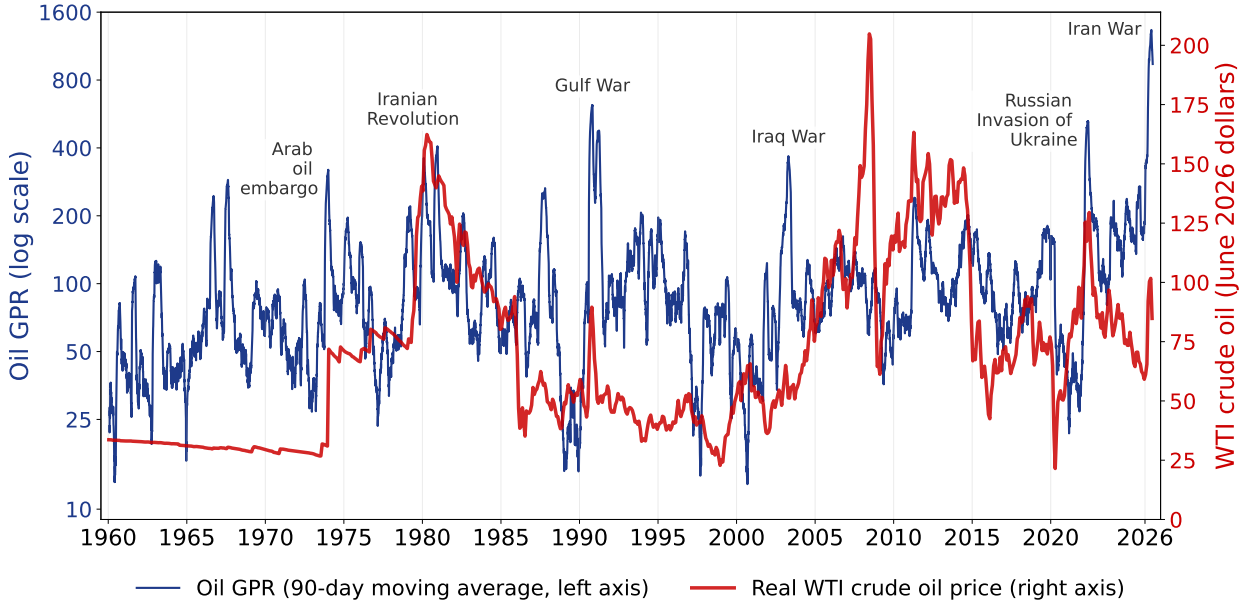
where $D_t$ is the set of articles on date $t$ classified as discussing a geopolitical oil disruption, $S_{it}$ is the article's GPR score, and $A_t$ is the total number of newspaper articles published on date $t$. The numerator weights each disruption article by its GPR intensity. Regional Oil GPR indexes are constructed similarly, restricting $D_t$ to articles classified as affecting a particular region. We also split the index along the threat-act dimension of Section 5.1: keeping only the disruption articles that the classifier labels as threats, or as both threats and acts, gives an Oil GPR threats index, and the complementary selection gives an Oil GPR acts index. The Oil threats index isolates the risk of a future disruption from the disruption itself, which is what makes it useful as an instrument for oil price movements in Section 6.2.

Figure 5 illustrates the Oil GPR index alongside oil prices. The index closely tracks the major geopolitical risk events that affected the oil market: the 1973 Arab oil embargo, the 1979 Iranian Revolution, the 1990–91 Gulf War, and the Russia–Ukraine conflict and associated sanctions from 2022 onward. The co-movement between the Oil GPR index and oil prices confirms that the LLM-based classification is capturing geopolitical episodes associated with significant oil market disruptions. A regional, time-series decomposition of the Oil GPR index by geography is provided in Appendix A.8.

Of all articles with a GPR score above 0.5, roughly 13 percent contain oil-related keywords and are passed to the second classification layer. Of these, roughly three-quarters are classified as discussing a geopolitical oil supply disruption. In the aggregate, the second classification layer matters little: the Oil GPR index has a correlation of 0.98 at monthly frequency with an index built from the keyword filter alone. However, the layer matters for what the keyword filter cannot

[21] To improve classification accuracy, the oil and country classifiers read the first 3,000 characters of each article, compared with 2,000 characters for the baseline scoring step and the threats classifier.

Figure 5: Oil GPR Index and Real Oil Prices




*Notes*: The blue line (left axis) is the Oil GPR index (90-day moving average of the daily series). The red line (right axis) is the monthly real WTI crude oil price, in June 2026 dollars. Source: authors' calculations; oil prices and consumer prices from FRED. Last observation: June 2026.

provide: the region affected by the disruption, and its nature (threats vs. acts). The Middle East accounts for nearly two-thirds of all disruption mentions, consistent with its dominant role in geopolitical oil risk throughout the sample period.

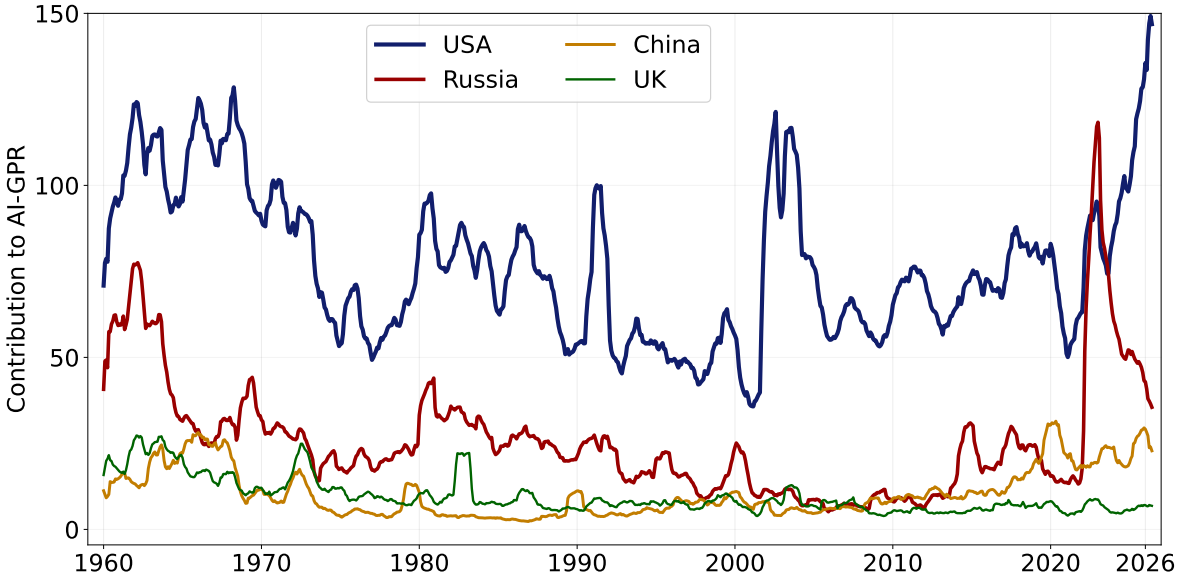
### 5.3 Country Risk and Bilateral Risk

Our final set of measures focuses on country-level and bilateral geopolitical risk.[22] The country index measures news coverage of adverse geopolitical events associated with a particular country. We pass every article receiving a score greater than 0 in the first stage through a second classification layer. This layer identifies which countries are involved in the geopolitical event and the role each plays: the initiator, which launches or triggers the hostile action; the respondent, which is its direct target; or a spillover party, which is not directly involved but is drawn in or affected, such as a neighboring country exposed to a conflict. We aggregate the AI-GPR scores for each country across all roles and scale to the overall index.

Figure 6 presents monthly series for the United States, Russia, China, and the United Kingdom—the four countries most often featured in newspaper coverage of geopolitical risks. The companion webpage includes time series of geopolitical risk for 200 countries. Because the index measures

[22] Country-level decompositions of this kind extend the country subindexes of Caldara and Iacoviello (2022), who constructed keyword-based series for selected nations.

Figure 6: AI-GPR by Country




*Notes*: Monthly AI-GPR contribution for the four most prominent countries across all actor roles (initiator, respondent, spillover), smoothed with a 12-month moving average. Each series is the country's contribution to the overall AI-GPR index, which has a mean of 100 over 1985–2019. Source: authors' calculations. Last observation: June 2026.

coverage, its level reflects both the severity of events and the attention that U.S. newspapers pay to a country. A coup in Rwanda receives less coverage than a naval standoff near Taiwan, and coverage of adverse events involving China grows with its economic and military footprint. The country indexes are therefore better suited to tracking how risk involving a given country changes over time than to comparing levels across countries.

The United States is strongly present throughout the entire sample, as an initiator, respondent, or spillover party in conflicts worldwide.[23] Russia registers two distinct elevated periods, once during the Cold War era through the late 1980s, and again during the post-2014 period following the annexation of Crimea through the full-scale invasion of Ukraine in 2022. China's GPR contribution rises steadily from the late 1990s onward, consistent with its growing global presence and the intensification of geopolitical competition in the Indo-Pacific.

We build on the country index to create bilateral indexes. Each index tracks geopolitical risk between an initiator and a respondent. We construct bilateral indexes for the top 1,200 country pairs as the sum of AI-GPR scores in articles where the first country is classified as initiator and the second as respondent, so that the series for the United States acting against Russia and for

[23] The large coverage of U.S.-related events also reflects our choice of U.S. newspapers. For measures built using our approach from non-U.S. news sources, see Ambrocio et al. (2025), Andres-Escayola et al. (2022), and Bondarenko, Lewis, Rottner, and Schüler (2024).

Russia acting against the United States are distinct.[24]

Assigning roles is where the LLM does what keywords cannot. A keyword search can tell that Russia and the United States both appear in an article. It cannot tell whether Russia is acting against the United States, whether the United States is responding to Russian pressure, or whether the two countries merely turn up in unrelated paragraphs. The LLM reads the article and records which country initiates the action and which is its target.

Figure 7 presents the nine country pairs with the highest average bilateral intensity, each shown in both directions. The thick blue line is the more intense direction, named in the panel title, and the thin red line is the reverse. The United States is the main initiator in eight of the nine pairs, reflecting both its central role in postwar geopolitics and the U.S. newspapers behind the index. The pairs span the Cold War confrontation with the Soviet Union and Cuba, the major U.S. military interventions in Vietnam, Iraq, and Afghanistan, tensions with China, Iran, and Israel, and Russia's 2022 invasion of Ukraine. When a large power confronts a smaller one, as in Iraq, Afghanistan, or Ukraine, coverage casts one country overwhelmingly as the initiator, while the reverse series moves very little. Between the United States and Russia, by contrast, initiation is split far more evenly across the two directions. Because the index tracks news coverage, it can fall while a conflict continues: the Russia–Ukraine series peaks with the 2022 invasion and by 2026 is a fraction of that peak, although the war has not ended.

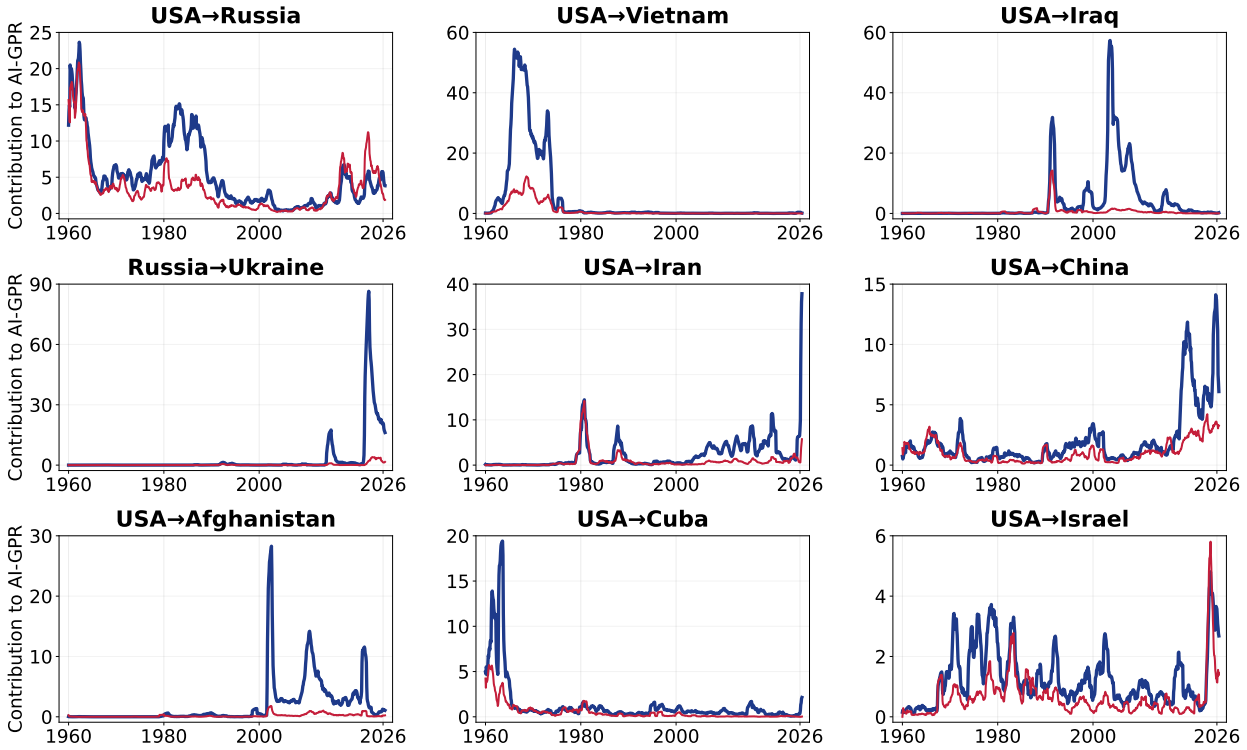
The same classification layer also labels each article by the type of geopolitical event it describes, including military conflict, diplomatic tension, terrorism, civil war, nuclear threat, coup or regime change, and sanctions. The resulting event-type series are available on the companion webpage. These categories track familiar historical patterns. Military conflict peaks around the Vietnam War, the Gulf War, and Russia's 2022 invasion of Ukraine. Terrorism rises sharply after the September 11 attacks. Sanctions become more prominent after 2014, as economic coercion becomes a favored geopolitical instrument.

# 6 Applications: Geopolitical Risk in Financial Markets, Oil Markets, and Trade

We present a series of applications relating the three dimensions of geopolitical risk to financial markets, oil markets, and international trade. First, we examine how geopolitical threats and acts are priced in the stock market using the threats and acts subindexes. We then study the macroeconomic response to geopolitical oil market shocks identified with an instrument constructed from the Oil GPR index. Lastly, we examine the dynamic response of international trade to bilateral

[24] Each directed-pair index is scaled like the country indexes, so the bilateral series share the denominator and scale of the aggregate index. Because an article in which one country acts against several respondents counts toward each pair, the bilateral indexes do not sum to the aggregate index.

Figure 7: AI-GPR Bilateral Indexes: Top Country Pairs, Both Directions




*Notes*: Each panel shows both directed bilateral AI-GPR indexes for a country pair (12-month moving average). The thick blue line is the more intense direction, named in the panel title as initiator → respondent, and the thin red line is the reverse. Each directed index sums the AI-GPR scores of articles in which the first country is classified as initiator and the second as respondent, and is the pair's contribution to the overall AI-GPR index, which has a mean of 100 over 1985–2019. The nine pairs have the highest average intensity in their dominant direction over the sample. Source: authors' calculations. Last observation: June 2026.

geopolitical risk.

## 6.1 Financial Markets and Geopolitical Threats and Acts

A large literature studies whether geopolitical events depress stock markets. We revisit this question by relating weekly excess returns on the U.S. stock market to changes in the AI-GPR index from 1960 to March 2026.[25] Specifically, we estimate:

$$r_t^e = \alpha + \beta\, \Delta\text{GPR}_t + \varepsilon_t \tag{3}$$

where $r_t^e$ is the weekly excess market return in percent and $\Delta\text{GPR}_t$ is the standardized weekly change in the AI-GPR. Table 2 reports the results. A one-standard-deviation rise in the AI-GPR

[25] Excess returns are from the Fama-French daily factors through November 2025, extended with the S&P 500 excess return from Haver Analytics. The GPR on each trading day averages the index over that day and any preceding days the market was closed, so news from weekends and holidays is assigned to the next trading day.

Table 2: Weekly Stock Returns and Geopolitical Risk

| | Aggregate GPR | | Threats and Acts | |
|---|---|---|---|---|
| | OLS | Decomposition | OLS | Decomposition |
| | (1) | (2) | (3) | (4) |
| $\Delta\text{GPR}_t$ | $-0.136^{**}$ | | | |
| | (0.057) | | | |
| $\Delta\text{GPR}_{t,\,threats}$ | | | $-0.081^{**}$ | |
| | | | (0.037) | |
| $\Delta\text{GPR}_{t,\,acts}$ | | | $-0.084$ | |
| | | | (0.061) | |
| Persistent $\widehat{\Delta\text{GPR}}_t$ | | $-0.271^{**}$ | | |
| | | (0.134) | | |
| Persistent $\widehat{\Delta\text{GPR}}_{t,\,threats}$ | | | | $-0.220^{**}$ |
| | | | | (0.103) |
| Persistent $\widehat{\Delta\text{GPR}}_{t,\,acts}$ | | | | $-0.063$ |
| | | | | (0.131) |
| Shock $\hat{u}_t$ | | $-0.119^{*}$ | | |
| | | (0.062) | | |
| Shock $\hat{u}_{t,\,threats}$ | | | | $-0.052$ |
| | | | | (0.044) |
| Shock $\hat{u}_{t,\,acts}$ | | | | $-0.090^{**}$ |
| | | | | (0.042) |
| Constant | $0.137^{***}$ | $0.138^{***}$ | $0.136^{***}$ | $0.138^{***}$ |
| | (0.037) | (0.037) | (0.037) | (0.037) |
| Observations | 3,452 | 3,448 | 3,452 | 3,448 |
| $R^2$ | 0.004 | 0.004 | 0.004 | 0.004 |
| Sample | 1960–2026 | 1960–2026 | 1960–2026 | 1960–2026 |

*Notes*: Weekly regressions of excess market returns ($r_t^e$, in percent) on standardized changes in the AI-GPR index. Columns (1)–(2) use the AI-GPR index; columns (3)–(4) use its threats and acts subindexes. Weekly stock returns are the sum of daily excess returns within the week, and the weekly GPR variable is the change in the within-week average of the trading-day GPR measure, standardized over the full sample. Columns (1) and (3) report OLS regressions of $r_t^e$ on the GPR change(s). Columns (2) and (4) decompose each GPR change into a persistent component (fitted values from an AR(4) model) and a shock component (residuals). Excess market returns are from the Fama-French daily factors dataset through November 28, 2025, extended with Haver SP500 excess returns (SP500 simple return minus the daily-compounded 3-month T-bill rate) from December 1, 2025 onward. OLS columns report heteroskedasticity-robust standard errors in parentheses; decomposition columns report pairs-bootstrap standard errors (1,000 replications). The estimation sample runs through March 2026. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

over a week is associated with a decline of about fourteen basis points in the weekly market return, about the average weekly equity premium over the sample.

The OLS estimate in equation (3) conflates predictable and unpredictable movements in GPR. To disentangle these, we decompose geopolitical risk into anticipated and unanticipated compo-

nents. We estimate an AR(4) model for the standardized GPR change:

$$\Delta\text{GPR}_t = \phi_0 + \sum_{j=1}^{4} \phi_j \, \Delta\text{GPR}_{t-j} + u_t, \tag{4}$$

and decompose $\Delta\text{GPR}_t$ into the persistent (predictable) component $\widehat{\Delta\text{GPR}}_t$ and the shock (unpredictable) component $\hat{u}_t$. We then estimate:[26]

$$r_t^e = \alpha + \beta_1 \, \widehat{\Delta\text{GPR}}_t + \beta_2 \, \hat{u}_t + \varepsilon_t. \tag{5}$$

We estimate both the aggregated $\Delta\text{GPR}_t$ regression and the decomposed $\Delta\text{GPR}_t$ twice, first using the aggregate AI-GPR index and then with the threat and act subindexes entered jointly. The two pairs of regressions map onto the four columns of Table 2: the aggregate index in columns (1) and (2), and its threat-and-act decomposition in columns (3) and (4). For the aggregate index, the predictable component of geopolitical risk depresses returns more than twice as much as the unexpected component; both coefficients are statistically significant, the unexpected one at the 10 percent level. A sudden geopolitical event like the September 11 attacks generates an immediate negative return, but the persistence of elevated risk in the weeks that follow continues to depress returns most. Markets re-price as risk persists and enters the outlook, not only on the day of the event.

The gap between the muted aggregate OLS estimate and the larger effect of the persistent component suggests that a simple linear regression may understate the role of geopolitical risk. The aggregate OLS coefficient may be attenuated by the high-frequency noise of weekly shocks, which are large in magnitude but transient in their effect on asset prices. This result echoes Gonçalves et al. (2025), who find that only the threat component of the original GPR index is consistently priced across assets.

Separating the index into its threat and act components further refines the channels through which geopolitical risk affects stock prices. Columns (3) and (4) show that the effect of threats on returns is precisely estimated, while acts are not statistically significant despite being comparable in magnitude. Additionally, the decomposition into anticipated and unanticipated movements shows that threats affect returns through their predictable component. As a threat persists, markets re-price gradually, so it is the slow-moving, anticipated component of threat risk that weighs on returns. By contrast, only the unanticipated occurrence of an act moves markets. This asymmetry also explains why acts look insignificant in the simple regression: pooling a surprise effect with a negligible anticipated one pushes the average toward zero.

[26] This approach follows the empirical tradition of Barro (1979), who decomposes macroeconomic variables into anticipated and unanticipated components to test whether agents respond differently to predictable trends versus stochastic innovations.

The improved day-to-day precision of the AI-GPR index is central to these results. When we repeat the same analysis using the keyword-based GPR index,[27] the results are qualitatively similar but economically and statistically weaker: the contemporaneous relationship with returns is roughly a third smaller and only marginally significant, and in the decomposition the persistent component barely clears conventional significance while the shock component loses it altogether. Appendix A.6 reports these regressions in full. These differences are consistent with attenuation bias: the keyword index misclassifies more articles, so more of its variation is unrelated to geopolitical risk.

### 6.2 Oil Market Shocks and their Macroeconomic Effects

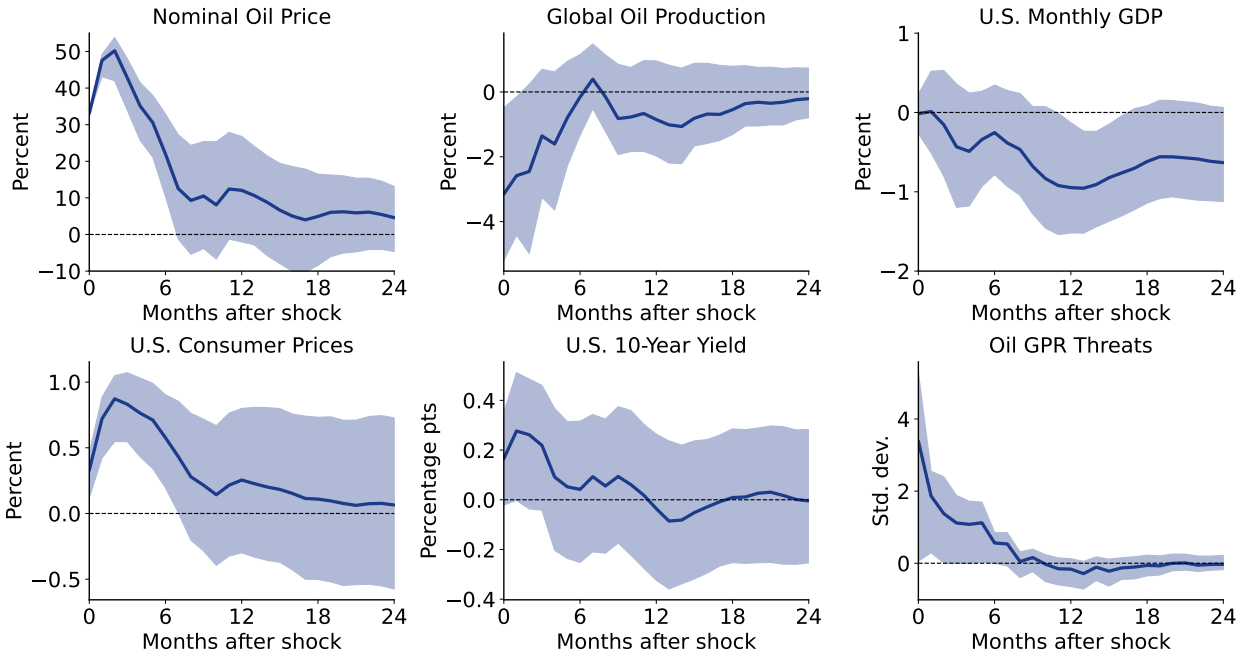
The Oil GPR index can help isolate oil price movements driven by geopolitical supply disruptions, as distinct from demand-driven fluctuations or weather-related supply shocks. We borrow this idea from Verduzco-Bustos and Zanetti (2026), who identify geopolitical oil market shocks from oil price changes on days with large jumps in the keyword-based GPR threats index. We replace their news series with our novel Oil GPR index, which isolates geopolitical news related to energy supply disruptions, and estimate a six-variable proxy structural VAR (Mertens and Ravn, 2013; Stock and Watson, 2012). The system includes the nominal U.S. crude oil price, global oil production, U.S. monthly GDP (the Brave-Butters-Kelley index, Brave et al. 2019), the U.S. consumer price index, the 10-year Treasury yield, and the Oil GPR threats index described below. The data are monthly and run from January 1985 to June 2026, and the VAR has 12 lags.[28]

Oil prices also respond to the state of the economy, so their movements cannot be read directly as supply shocks, and the proxy VAR needs an external instrument that moves oil prices through supply news alone. Our instrument measures how oil prices move on days when the news is dominated by threats to oil supply. We select the days on which the threat component of the Oil GPR index is more than one standard deviation above its mean, and compute the change in the log WTI spot price from the trading day before to the trading day after. Because these days are chosen on the basis of the news rather than prices, and global demand conditions change little over such a short window, the price changes mainly reflect news about future oil supply. Summing them within each month gives the instrument, which has a first-stage $F$-statistic of 38.[29] In August 1990, the month Iraq invaded Kuwait, oil prices rose by about a third, and the days flagged by the Oil GPR account for most of that rise.

Figure 8 presents the impulse responses to a geopolitical oil market shock, normalized to raise oil prices by about one third on impact and 50 percent at the peak. The impact effect is roughly the increase after Iraq invaded Kuwait in August 1990, and about four standard deviations of the

[27] We compute the keyword-based index on the same three newspapers as the AI-GPR; see Appendix A.5.
[28] Daily oil price quotes start in 1986, so the instrument is zero for every month of 1985.
[29] Appendix A.9 reports the details and plots the instrument against the monthly change in oil prices. We shift each flagged date back by one day because newspapers report events the day after they happen.

Figure 8: Impulse Responses to a Geopolitical Oil Market Shock




*Notes*: Impulse responses to a geopolitical oil market shock identified with a proxy structural VAR. The shock is normalized to raise oil prices by 33 log points on impact. Shaded areas are 80 percent confidence intervals from a moving block bootstrap following Jentsch and Lunsford (2022). Source: authors' calculations.

shock. Oil prices keep rising for two months and then fall back gradually. The Oil GPR threats index jumps on impact and stays elevated for about six months. Oil production falls on impact before gradually recovering. Consumer prices rise, peaking at close to 1 percent above baseline, and 10-year yields rise. U.S. GDP declines gradually and is about 1 percent lower at the trough, a year after the shock, reflecting the contractionary effect of higher energy costs as well as the broader disruptions that accompany geopolitical events affecting the oil market.

### 6.3 Bilateral Geopolitical Shocks and International Trade

We ask whether a rise in geopolitical tensions between two countries is followed by a decline in trade between them, as firms reroute supply chains, governments impose sanctions, and shipping gets riskier. We merge our bilateral AI-GPR index with the CEPII Gravity database (Conte et al., 2023), which assembles UN Comtrade flows into a harmonized dataset of annual bilateral trade. Our variable of interest is total two-way real merchandise trade for each country pair, obtained by summing the two directed flows (in U.S. dollars) and deflating using the U.S. GDP deflator.

The trade data cover the years from 1962 through 2020.[30] Because total trade has no direction, we average the two directed bilateral indexes of each pair into a single series. The directed series, which separate the initiator from the target, are available on the companion webpage for applications where direction matters.

Trade flows are likely to respond to geopolitical news with some delay, as cross-country relationships take time to build and redirect. Fan et al. (2026), who construct a bilateral geopolitical alignment score from a large-scale classification of political events, find that trade adjusts to changes in alignment gradually over several years. A static regression of trade on contemporaneous geopolitical risk may thus miss these effects. To trace the dynamic response of trade to bilateral geopolitical risk at each horizon, we estimate:

$$y_{od,t+h} - y_{od,t-1} = \alpha_{od,h} + \alpha_{o,t,h} + \alpha_{d,t,h} + \gamma_h \, Z_{od,t} + \mathbf{d}_h \mathbf{X}_{od,t} + \varepsilon_{od,t+h}, \tag{6}$$

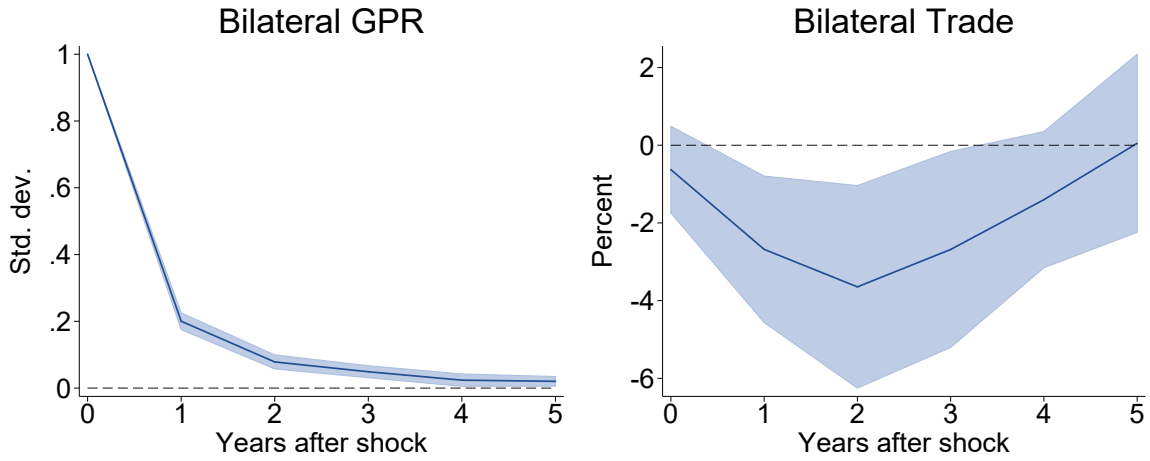
where $y_{od,t} = 100 \times \log Y_{od,t}$, $Y_{od,t}$ is total trade between countries $o$ and $d$, summing the flows in both directions, and $Z_{od,t}$ is the bilateral AI-GPR for the pair, standardized within country pair so that it measures how far a relationship has moved relative to its own long-run average. For each horizon $h$, the coefficient $\gamma_h$ is the percent response of bilateral trade to a one-standard-deviation increase in bilateral geopolitical risk. The regression includes country-pair fixed effects ($\alpha_{od}$) and year dummies for each of the two countries ($\alpha_{o,t}$ and $\alpha_{d,t}$). Controls $\mathbf{X}_{od,t}$ include three lags of bilateral GPR, three lags of $y_{od,t}$, and a dummy for whether the pair is party to a regional trade agreement.[31] Conditioning on past GPR is what makes $Z_{od,t}$ a shock rather than a level. Standard errors are clustered by each of the two countries in the pair and by year, since geopolitical events move many country pairs in the same year. We estimate the same projection, with the same right-hand side, replacing trade with bilateral GPR on the left.

Figure 9 plots the results. A typical bilateral shock raises the index only briefly. Part of the decline reflects newspapers writing less about a conflict once it is no longer new, even while it continues, as the Russia–Ukraine series in Figure 7 shows. The right panel shows that trade responds more slowly. It barely moves on impact, then declines over the following two years, reaching a trough of almost 4 percent below baseline before gradually recovering. Trade is still declining at horizons where coverage of the risk has largely faded, which is what one would expect if contracts and supply relationships take time to redirect. This transitory decline in trade is consistent with the evidence in Fan et al. (2026) for a transitory shock to geopolitical alignment.

[30] Year 2020 is the last year in the current release of the database. Comtrade records each flow twice, as reported by the exporter and by the importer; we use the importer's report and fall back on the exporter's where it is missing.

[31] The variable is taken from the WTO's Regional Trade Agreements database, supplemented by Thierry Mayer, as distributed in Conte et al. (2023).

Figure 9: Response of Bilateral GPR and Trade to a Bilateral GPR Shock




*Notes*: Responses to a one-standard-deviation increase in bilateral GPR, where the standard deviation is computed within country pair. Both panels plot the value at $t+h$ minus the value at $t-1$. The left panel is bilateral GPR, in standard deviations. The right panel plots the coefficients $\gamma_h$ from equation (6), the percent response of total two-way bilateral trade. Shaded areas are 90 percent confidence intervals, with standard errors clustered by each of the two countries in the pair and by year. Source: authors' calculations.

# 7 Conclusion

We have introduced a set of measures of geopolitical risk built by having a large language model read newspaper articles. The headline AI-GPR index keeps the definition of geopolitical risk behind the original GPR but scores each article on a continuous scale, and it misses fewer relevant articles and flags fewer irrelevant ones than a keyword search. From the same articles we build measures that keyword counts deliver only roughly or not at all: threats and acts, geopolitical risks to oil supply by region, geopolitical risk for two hundred countries, and risk directed from one country to another. All of them are updated monthly on the companion webpage.

We illustrated several applications of these measures. First, the AI-GPR produces a more precisely estimated negative effect of geopolitical risk on stock returns and shows that markets are especially sensitive to the persistent component of GPR innovations. Further decomposing the index into threats and acts shows that this persistent effect runs through anticipated threats, while the surprise component of realized acts also weighs on prices. Second, by asking the model which oil supplies are at risk, we constructed a historical time series of geopolitical oil supply disruptions by region, showing the shifting geography of energy-related geopolitical risk over six decades and the macroeconomic effects of geopolitical oil market shocks. Third, we extract the geopolitical actors in each event and their roles. This yields country indexes, which extend the country subindexes of the original GPR, and indexes for directed country pairs, which identify who is acting on whom rather than which countries merely appear together. A rise in bilateral GPR is followed by a decline in

trade between the two countries.

Our main contribution is demonstrating that LLM-based approaches can be applied at scale to construct consistent historical time series spanning multiple decades and expanding the information content of indexes based on simpler keyword-based approaches. As computational costs decline, this becomes practical for a wide range of text-based measurement in economics. Future research could extend it to firm-level or sector-level measures of geopolitical risk exposure, or use the bilateral and country-level indexes to study how geopolitical risks transmit across countries and markets.

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# A Appendix

## A.1 Alternative Versions of AI Model to Classify Articles

We classify a sample of 2,050 articles using four OpenAI models—GPT-4o mini, GPT-4o, GPT-5-mini, and GPT-5—together with the open-weight Llama 3.1 8B Instruct, all with the same prompt. We set temperature to zero for the models that support it; the GPT-5 models do not expose an adjustable temperature and run at their default settings. Table A.1 reports the full pairwise correlation matrix.

Table A.1: Cross-Model Score Correlations

| | GPT-4o mini | GPT-4o | GPT-5-mini | GPT-5 | Llama |
|---|---|---|---|---|---|
| GPT-4o mini | 1.00 | | | | |
| GPT-4o | 0.87 | 1.00 | | | |
| GPT-5-mini | 0.87 | 0.86 | 1.00 | | |
| GPT-5 | 0.87 | 0.88 | 0.94 | 1.00 | |
| Llama 3.1 8B | 0.91 | 0.85 | 0.84 | 0.84 | 1.00 |

*Notes*: Pairwise Pearson correlations of geopolitical risk scores assigned by each model to a sample of 2,050 newspaper articles drawn from all three publications. All models use the same classification prompt, at temperature zero where supported; the GPT-5 models run at their default settings. "Llama" denotes the open-weight Llama 3.1 8B Instruct model.

All pairwise correlations are high, with the closest agreement between GPT-5-mini and GPT-5. The mean score varies modestly across models, reflecting differences in calibration that wash out after normalization.

## A.2 Alternative Ways of Constructing the Index

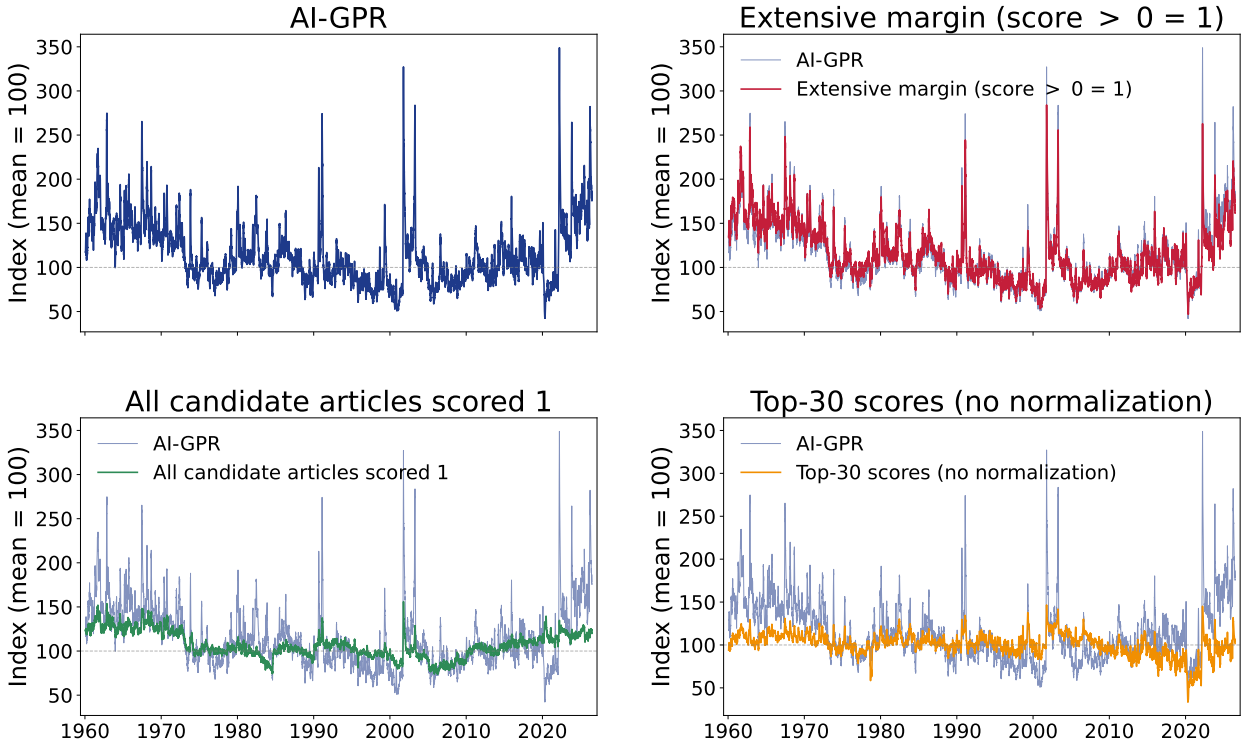
The baseline index in equation (1) makes two choices: it weights each article by its graded risk score $S_{it}$ (the intensive margin), and it normalizes the score sum by total newspaper output $A_t$ rather than by the number of scored articles $N_t$. To show the role of these choices, we recompute the index under four alternative constructions, each reindexed to a mean of 100 over 1985–2019: (i) the baseline AI-GPR, $\sum_i S_{it}/A_t$; (ii) an extensive-margin version in which every article with $S_{it} > 0$ counts as one, $\#\{i : S_{it} > 0\}/A_t$, discarding intensity; (iii) an all-candidate version in which every scored article counts as one regardless of its score, equal to the scored share $N_t/A_t$; and (iv) an unnormalized version equal to the sum of the thirty highest scores each day, $\sum_{\text{top }30} S_{it}$, with no denominator.

Figure A.1 plots the four series and Table A.2 reports their pairwise correlations. The baseline and the extensive-margin version are very close (correlation 0.97): the intensity weighting is not what generates the index's movements, although it emphasizes some of the spikes. Replacing the graded score sum in the numerator with a simple count of scored articles—the all-candidate share $N_t/A_t$—lowers the correlation to 0.70, and removing the denominator $A_t$ entirely (the unnormalized top-30 sum) gives 0.48 in levels; both are more closely related to the baseline in year-on-year changes (0.66 and 0.85, respectively), so these construction choices matter more for the level of the index than for its changes.

The third construction checks for long-run drift in the scored share $N_t/A_t$, as newspaper composition and the mix of topics shift over six decades. As Figure A.1 shows, the share of candidate

articles is a smooth, slowly drifting series with none of the event spikes that define the geopolitical risk index, and among the four it least resembles the baseline's sharp, event-driven movements. In addition, the series shows a U-shape that mostly reflects the inverted U-shape in the pattern of total newspaper articles over the sample. The index's economically relevant variation—the sharp rises around the Gulf War, September 11, the Iraq War, and Russia's invasion of Ukraine—comes from the geopolitical content of the articles, not from how many articles clear the generous keyword filter.

Figure A.1: Alternative Versions of the Index




*Notes*: Four alternative constructions of the AI-GPR index, each shown as a 30-day moving average and reindexed to a mean of 100 over 1985–2019. AI-GPR is the baseline $\sum_i S_{it}/A_t$; Extensive margin counts each article with $S_{it} > 0$ as one; All candidate articles scored 1 counts every scored article as one, equal to $N_t/A_t$; Top-30 scores sums the thirty highest scores each day with no normalization. Source: authors' calculations. Last observation: June 2026.

## A.3 Additional Details on the Human Audit

This appendix gives the full results of the human validation summarized in Section 4.1. We hand-score 2,447 articles drawn from the *New York Times*, the *Washington Post*, and the *Chicago Tribune* between January 1960 and December 2025. We use the same prompt and guidelines that the model uses to score the articles. Of these, 2,101 were scored by two annotators, and we take the consensus of the available coders as the human score and compare it with the LLM's score.

The audit is stratified rather than a simple random draw, because geopolitical articles are a small minority of the corpus and a proportional sample of the corpus would contain unrelated articles not flagged by our filters. Table A.3 gives the design. Stratum C, articles that are flagged

Table A.2: Correlations Across Index Constructions

| | (1) | (2) | (3) | (4) |
|---|---|---|---|---|
| (1) AI-GPR (baseline) | 1.00 | | | |
| (2) Extensive margin | 0.97 | 1.00 | | |
| (3) All candidate ($N_t/A_t$) | 0.70 | 0.75 | 1.00 | |
| (4) Top-30 (unnormalized) | 0.48 | 0.41 | 0.08 | 1.00 |

*Notes*: Pairwise Pearson correlations of the four index constructions defined in the text, computed on the daily series (reindexed to a mean of 100 over 1985–2019) over 1960–2026.

by the keyword filter and which the model scored above zero, is where humans can disagree on an article the LLM scored positively; stratum B, flagged but scored zero, is where the model's misses would show up; and stratum A, never retrieved at all, is where the keyword filter's misses would live. The three strata make up the corpus, so weighting each by its population share recovers rates for the full sample of 8.5 million articles. The weights differ substantially—an audited article in stratum A stands for five times as many corpus articles as one in stratum C—so the pooled, unweighted rates describe the audited panel and not the corpus, and we report both.

Table A.3: Design of the Human Audit

| Stratum | Share of corpus (%) | Articles audited | Corpus articles each represents |
|---|---|---|---|
| A. No keyword match | 46.3 | 400 | 9,893 |
| B. Keyword match, AI score 0 | 40.0 | 1,451 | 2,357 |
| C. Keyword match, AI score $> 0$ | 13.7 | 596 | 1,970 |
| Total | 100.0 | 2,447 | 3,495 |

*Notes*: The three strata partition the 8.5 million articles published in the three newspapers over 1960–2025. Stratum A is articles the keyword pre-filter never sent to the LLM, drawn from the same pool as the false-omission check of Section 4.2; strata B and C are articles it did send, split by whether the model scored them above zero. The last column is the stratum population divided by the number of articles audited in it, that is, the weight each audited article carries when rates are computed for the corpus.

The human coders agree with one another imperfectly: on the double-coded articles in strata B and C their scores correlate at 0.78 and concur on whether an article carries any geopolitical risk in about 92 percent of cases. Even careful readers applying the same rubric make different judgment calls, which sets a benchmark for any classifier. Against the human consensus, the machine scores correlate at 0.85—above the coders' agreement with each other—agree on the presence of any geopolitical risk in about 90 percent of cases, and have a mean absolute difference of about 0.05, roughly the spacing of a single step on the scale. Table A.4 reports the full cross-tabulation; the off-diagonal mass sits almost entirely on the first off-diagonal, so disagreements are overwhelmingly one-step judgment calls at a rubric boundary, such as whether an ongoing regional conflict constitutes "significant tensions" (0.4) or "major war escalation risks" (0.6). The model's and the humans' average scores are essentially identical (both 0.122), so the article-level disagreements are offsetting and introduce no aggregate bias in levels.

The model's disagreements with the humans are, moreover, concentrated in the articles the

humans themselves find ambiguous. On the double-coded articles where the two coders assign identical scores, the model matches their common score exactly in about 90 percent of cases, with a mean absolute deviation of 0.02; where the coders disagree with each other, the model's mean absolute deviation from the consensus rises to 0.14. Model error thus tracks genuine ambiguity in the articles rather than occurring at random, the same pattern Hartley (2025) documents for policy-uncertainty classification, where model misclassification falls by half as human-coder certainty rises.

Table A.4: Human vs. AI Score Cross-Tabulation

| | AI score | | | | |
|---|---|---|---|---|---|
| Human score | 0.0 | 0.2 | 0.4 | 0.6 | 0.8 |
| 0.0 | 1366 | 60 | 23 | 3 | 0 |
| 0.2 | 43 | 34 | 47 | 6 | 0 |
| 0.4 | 39 | 60 | 140 | 101 | 0 |
| 0.6 | 3 | 0 | 21 | 73 | 5 |
| 0.8 | 0 | 1 | 3 | 15 | 4 |

*Notes*: Cross-tabulation of the human consensus score (rounded to the nearest 0.2) and the GPT-4o mini score, for the 2,047 audited articles in strata B and C of Table A.3. The 400 stratum-A articles are left out because the pre-filter never sent them to the model, so they carry no machine score in the index. Both human coders scored 388 of the 400 zero, one coder found geopolitical content in two and the other in twelve, and no article was scored above 0.4. Scored for the false-omission check of Section 4.2, the model flags one of them, the article both coders flag, at the same 0.2. These articles enter the corpus-weighted rates of Table A.5. Rows are human scores, columns are model scores. The model never assigned a score above 0.8. Off-diagonal mass is concentrated on the first off-diagonal, that is, one-step disagreements.

Scoring the same audited articles with both classifiers also allows a like-for-like comparison against the human labels, free of the differences in sample, rubric, and ground truth that complicate cross-study figures. Table A.5 reports this three-way comparison, pooling the individual article–coder ratings in the two keyword-retrieved strata (3,748 pairs, with each double-coded article entering once per coder) and applying the original keyword query—exclusion words included—to the same texts. Benchmarking each coder's rating separately, rather than the consensus mean, ensures that an article one coder scores as zero and another as positive does not count as positive by construction. Within each method's own set of flagged articles, the share a human coder scores as zero is lower for the AI-GPR (a false-discovery rate of about 13 percent for the AI-GPR and 16 percent for the keyword query, the latter below the 21 percent reported in the original audit; reweighted to the corpus, the rates are 13 and 18 percent, the figures reported in Section 4.1), confirming the precision advantage discussed there. Measured against the consensus mean instead, the false-discovery rates are somewhat lower for both methods (9.4 and 15.1 percent), since averaging turns some single-coder zeros into positive benchmarks. The keyword query flags far fewer articles overall. Bringing stratum A back in and reweighting the three strata to their corpus shares, in the right-hand panel of the table, leaves that gap essentially intact: measured against the whole sample of 8.5 million articles rather than the retrieved strata alone, the AI-GPR fails to flag about a quarter of what a human coder scores as geopolitical, the keyword query more than four fifths. Stratum A is the one input the retrieved strata cannot supply, and the two coders who read it differ on how much geopolitical content it holds: one found it in half a percent of the articles, the other in three percent. Taking either coder alone instead of both moves the AI-GPR's corpus miss rate by about three percentage points in either direction and the keyword query's by less than one, so

the gap between the two methods does not depend on which reading one takes. The reweighting matters more for the keyword query's false-discovery rate, which rises by close to two percentage points because its flagged articles are drawn from both of the retrieved strata, while the AI-GPR's is unchanged because its flagged set is stratum C itself.

Table A.5: Human, AI-GPR, and Keyword GPR Compared

| | Retrieved strata (unweighted) | | | Full corpus (stratum-weighted) | | |
|---|---|---|---|---|---|---|
| | Flagged | False-disc. | Miss | Flagged | False-disc. | Miss |
| Human coders | 30.5 | — | — | 15.6 | — | — |
| AI-GPR | 29.2 | 13.2 | 16.9 | 13.7 | 13.2 | 23.8 |
| Keyword GPR | 6.9 | 16.3 | 81.1 | 3.3 | 18.1 | 82.5 |

*Notes*: The audit covers all 2,447 articles of Table A.3, stratum A included; the two panels differ in which of them they aggregate and how. The left-hand panel pools the article–coder ratings in the two strata the keyword filter retrieved, each article entering once for every human coder who scored it, 3,748 ratings in total across the *New York Times*, *Washington Post*, and *Chicago Tribune*. It sets stratum A aside because the three strata are sampled at different rates, so an unweighted pooling is interpretable only across strata sampled alike. The right-hand panel brings stratum A back in, where both coders read every article and each rating enters once, and weights the three strata by their corpus shares, so that it describes the full sample of 8.5 million articles. An article is flagged by a method when its score is positive; the keyword row applies the original GPR query, including its exclusion words, to the same texts. The false-discovery rate is the share of a method's flagged articles that the human scores as zero. The miss rate is the share of articles the human scores as positive that the method fails to flag. The AI-GPR's false-discovery rate is the same in both panels because its flagged set coincides with stratum C, while the keyword query's rises under reweighting because its flagged set spans strata B and C.

Reading the disagreeing articles and recording notes on each case, we group the discrepancies into five recurring patterns.

First, the model treats occasional economic and financial tensions between countries as geopolitical risks. Articles on a proposed U.S. tariff on Japanese goods (1971), the financial crisis in Latvia (2009), Iceland's refusal to repay British and Dutch lenders (2010), and strained Taiwan–China business ties (2000) were scored 0.4 to 0.6 by the model and 0 to 0.2 by us. The rubric defines geopolitical risk around war, terrorism, and military tensions. Trade and financial disputes between friendly nations strain "ties" in a literal sense but rarely carry risk of armed conflict, and a human coder discounts them accordingly. Many of the cases in which the model scores higher than the human are of this type, consistent with the model anchoring on surface vocabulary ("threat," "retaliation," "frayed ties") rather than the military nature of the underlying risk.

Second, the model treats domestic unrest and domestic security as geopolitical risk, even if the original definition emphasizes the interstate nature of such risks. Internal events with security overtones—the 1991 Crown Heights riots, labor strife framed around a general strike, an article on the disciplining of unruly airline passengers before the 2021 inauguration—were scored 0.4 to 0.6 by the model. The prompt explicitly instructs that domestic political events should score zero absent international implications, but the model appears to weight words such as "violence," "riot," and "security" heavily even in domestic settings.

Third, the model makes some minor temporal-attribution errors in both directions. The rubric instructs that retrospective coverage of old events should score zero unless current risks are men-

tioned. In some cases the model fails to recognize aftermath coverage as backward-looking: a 2004 article on legal turmoil in a Detroit terrorism prosecution (events eight months past) received 0.6, and a 2002 piece on routine airport security received 0.4. In other cases it applies the retrospective rule too aggressively: human-interest profiles published days after the 1993 World Trade Center bombing and the 1998 embassy bombings were scored 0 by the model even though the attacks were current events signaling elevated terrorism risk. The model evaluates the 2,000-character snippet in isolation, whereas a human coder knows the article's date and the surrounding history.

Fourth, the human coder sometimes credits context the snippet only implies. A 1961 wire item on a Navy missile contract was scored 0.2 by us (arms development at the height of the Cold War) and 0 by the model, which is arguably the more faithful reading of the prompt's instruction to classify based only on what an article states or strongly implies. A brief 2003 item on the capture of Ali Hassan al-Majid scored 0.6 in our coding—it presupposes the ongoing Iraq War—and 0 from the model. Reports on the humanitarian consequences of conflict, such as a 1973 piece on war-induced famine in Cambodia, scored higher in our coding because we treated humanitarian devastation and the implied risk of escalation as risk content while the model adhered to the narrower definition. In these cases the human is more generous than the prompt strictly licenses, so the disagreement reflects the rubric as much as the model.

Fifth, the model occasionally scores the backdrop rather than the subject of an article. When a major conflict serves as background for a story about something else—a 2023 piece on how the war in Ukraine affects livelihoods in a Norwegian border town, a 2017 color piece on the mood in a Beirut neighborhood—the model seems to score the conflict (0.6) while we score the article (0.2). The conflicts are real, but the articles convey little incremental information about war risk. These are occasional instances where a human coder weighs the subject of the article, while the model weighs the most alarming entity mentioned in it.

Finally, the model appears to compress the top of the scale: it rarely uses the highest gradations and almost never scores above 0.8, whereas human coders occasionally do. The compression is mild and affects few observations, but it implies that the machine index understates the relative intensity of the most extreme episodes. Taken together, the audit supports treating the LLM scores as a reliable proxy for careful human coding, with the caveat that the index seems to run slightly hot at the bottom of the scale, and slightly cold at the top.

The statistical correction of Carlson and Dell (2025) could in principle be applied to the AI-GPR, but two features of our audit would matter. Their framework requires a validation sample that is representative of the corpus (missing at random), whereas our audit oversamples articles that the model scores as geopolitically relevant, so the correction would need the random component of the audit or a reweighted version of it. The correction also inherits the noise in the human labels: the double-coding shows that two human coders agree with each other less closely than the model agrees with an individual coder, a limit on annotation quality that their framework takes as given.

## A.4 Hindsight Robustness Check

This appendix accompanies the hindsight discussion in Section 4.2. To test whether the model's knowledge of how historical events unfolded contaminates its scores, we re-score the articles around four major events with a date-anchored variant of the baseline prompt. The variant fixes each article's own publication date as the present, supplies the date, headline, and publisher alongside the article text, and instructs the model that it has no knowledge of anything that happened afterward. The full prompt is reproduced below; its geopolitical risk definition and 0.0–1.0 scoring scale are identical to the baseline prompt of Section 2.2.

```
HINDSIGHT-CHECK PROMPT (system message, prepended with the article's own date):

Today is {date}
You are a geopolitical risk analyst reading a news article.  You possess
absolutely zero knowledge of any events that occurred after today.
Classify the article's assessment of geopolitical risk based only on what the
article states or strongly implies.
[The geopolitical risk definition and the 0.0--1.0 scoring scale follow verbatim
from the baseline prompt in Section 2.2.]

User message:
Date:  {date}
Title:  {title}
Publisher:  {publisher}
{article text}
```

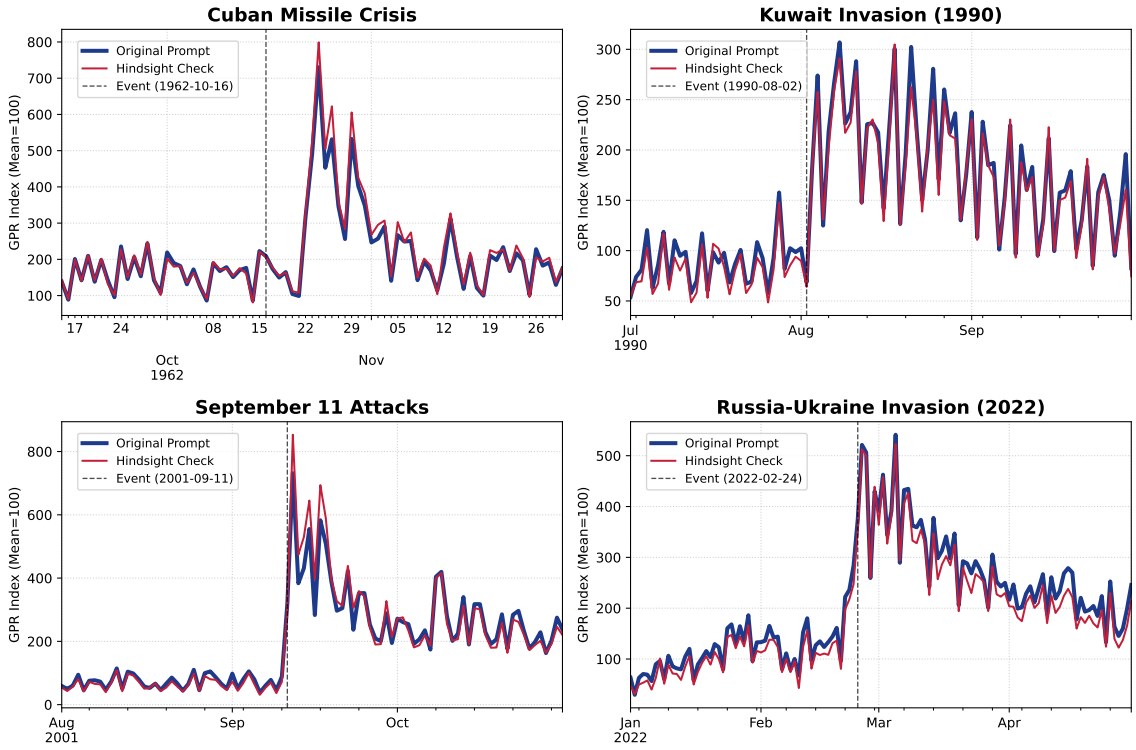
Figure A.2 plots the resulting daily series against the baseline around each event. The two are nearly indistinguishable. On average the hindsight-corrected scores are marginally higher for the Cuban Missile Crisis and September 11—the direction consistent with mild hindsight dampening in the baseline—and marginally lower for Kuwait and Russia–Ukraine, but the differences are very small throughout.

## A.5 The Original Keyword-Based GPR Index

As described in Section 3, we compare the AI-GPR index against the original, keyword-based GPR index developed by Caldara and Iacoviello (2022), using the same three newspapers and time period as the AI-GPR. In the original formulation, the search terms are organized into eight categories, each combining a geopolitical term group with a risk/threat term group using either the AND or the N/2 (words must appear within 2 words of each other) proximity operators. The full search string is:

```
KEYWORD-BASED GPR SEARCH QUERY:
(war OR conflict OR hostilities OR revolution* OR insurrection OR uprising OR
revolt OR coup OR geopolitical) N/2 (risk* OR warn* OR fear* OR danger* OR threat*
OR doubt* OR crisis OR troubl* OR disput* OR concern* OR tension* OR imminen* OR
inevitable OR footing OR menace* OR brink OR scare OR peril*)
OR (peace OR truce OR armistice OR treaty OR parley) N/2 (menace* OR reject* OR
threat* OR peril* OR boycott* OR disrupt*)
OR (military OR troops OR missile* OR ‘‘arms’’ OR weapon* OR bomb* OR warhead*)
AND (buildup* OR build-up* OR blockad* OR sanction* OR embargo OR quarantine OR
ultimatum OR mobiliz*)
OR (‘‘nuclear war’’ OR ‘‘atomic war’’ OR ‘‘nuclear missile*’’ OR ‘‘nuclear bomb*’’
OR ‘‘atomic bomb*’’ OR ‘‘h-bomb*’’ OR ‘‘hydrogen bomb*’’ OR ‘‘nuclear weapon*’’)
AND (risk* OR warn* OR fear* OR danger* OR threat* OR doubt* OR crisis OR troubl*
OR disput* OR concern* OR tension* OR imminen* OR inevitable OR footing OR menace*
OR brink OR scare OR peril*)
```

Figure A.2: Baseline and Hindsight-Anchored AI-GPR Around Key Events




*Notes*: Each panel plots the daily AI-GPR index around a major geopolitical event under the baseline prompt (blue, thick) and under the hindsight-check prompt (red, thin), which fixes the article's publication date as the present and instructs the model that it has no knowledge of subsequent events. Both series are scored on the same articles, normalized by total newspaper article count, and scaled to a mean of 100 over 1985–2019. Vertical dashed lines mark the event date. Source: authors' calculations.

```
OR (terroris* OR guerrilla* OR hostage*) N/2 (risk* OR warn* OR fear* OR danger*
OR threat* OR doubt* OR crisis OR troubl* OR disput* OR concern* OR tension* OR
imminen* OR inevitable OR footing OR menace* OR brink OR scare OR peril*)
OR (war OR conflict OR hostilities OR revolution* OR insurrection OR uprising OR
revolt OR coup OR geopolitical) N/2 (begin* OR begun OR began OR outbreak OR
''broke out'' OR breakout OR start* OR declar* OR proclamation OR launch*)
OR (allie* OR enem* OR foe* OR army OR navy OR aerial OR troops OR rebels OR
insurgen*) N/2 (drive* OR shell* OR advance* OR offensive OR invasion OR invad* OR
clash* OR attack* OR raid* OR launch* OR strike*)
OR (terroris* OR guerrilla* OR hostage*) N/2 (act OR attack OR bomb* OR kill* OR
strike* OR hijack*)
NOT (movie* OR film* OR museum* OR anniversar* OR obituar* OR memorial* OR arts OR
book OR books OR memoir* OR ''price war'' OR game OR story OR history OR veteran*
OR tribute* OR sport OR music OR racing OR cancer OR ''real estate'' OR mafia OR
trial OR tax)
```

Table A.6: Weekly Stock Returns: AI-GPR versus the Original Keyword-Based GPR

| | AI-GPR | | GPR-AER | |
|---|---|---|---|---|
| | (1) OLS | (2) Decomposition | (3) OLS | (4) Decomposition |
| $\Delta\text{GPR}_t$ | $-0.136^{**}$ | | $-0.097^{*}$ | |
| | (0.057) | | (0.053) | |
| Persistent $\widehat{\Delta\text{GPR}}_t$ | | $-0.271^{**}$ | | $-0.239^{**}$ |
| | | (0.134) | | (0.118) |
| Shock $\hat{u}_t$ | | $-0.119^{*}$ | | $-0.072$ |
| | | (0.062) | | (0.064) |
| Constant | $0.137^{***}$ | $0.138^{***}$ | $0.137^{***}$ | $0.138^{***}$ |
| | (0.037) | (0.037) | (0.037) | (0.038) |
| Observations | 3,452 | 3,448 | 3,452 | 3,448 |
| $R^2$ | 0.004 | 0.004 | 0.002 | 0.003 |
| Sample | 1960–2026 | 1960–2026 | 1960–2026 | 1960–2026 |

*Notes*: Weekly regressions of excess market returns on standardized changes in GPR, identical specification and sample to Table 2. Columns (1)–(2) use the AI-GPR index; columns (3)–(4) use GPR-AER, the reconstructed three-newspaper keyword index described in Section A.5. OLS columns report heteroskedasticity-robust standard errors in parentheses; decomposition columns report pairs-bootstrap standard errors (1,000 replications). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The resulting index is the ratio of articles matching the proximity search to total articles published on each date, normalized to have a mean of 100 over 1985–2019. This is the "GPR (Original)" series reported throughout the paper.

## A.6 Stock Returns Using the Original GPR Index

Section 6.1 reports that repeating the stock-return regressions with the keyword-based GPR index in place of the AI-GPR yields qualitatively similar but weaker results. Table A.6 reports the regressions underlying that comparison in full, alongside the AI-GPR results of Table 2 for reference. Columns (1)–(2) reproduce Table 2's aggregate-index columns exactly; columns (3)–(4) repeat the same specification with GPR-AER, the reconstructed three-newspaper index described in Section A.5. The aggregate OLS coefficient under GPR-AER is about a third smaller than under AI-GPR and only marginally significant. In the decomposition, the persistent component remains negative and barely clears the 5 percent level, while the shock component loses significance altogether—consistent with the view that the keyword index's higher misclassification rate attenuates the estimated relationship.

## A.7 Prompts for the AI-GPR Subindexes

We use the following prompts to classify articles for the GPR subindexes.

### A.7.1 Threats and Acts GPR

The threats and acts classification prompt identifies the nature of geopolitical risk. Relevant articles are labeled based on whether they predominantly discuss anticipated risk (threats), realized risk (acts), or both.

```
THREATS AND ACTS CLASSIFICATION PROMPT:
Analyze this newspaper article about geopolitical risk.  This article has been
identified as discussing an adverse geopolitical event (Geopolitical risk).
Geopolitical risk (GPR) encompasses adverse geopolitical events and associated
risks, including:
- Wars (nuclear and conventional)
- Terrorist threats and acts
- Military buildups and tensions
- International crises and conflicts
TASK: Classify the nature of geopolitical risk discussed in this article.
The three types of geopolitical risk nature are defined as follows:

  1. threat:  Risks, tensions, warnings, or potential escalations related to
     adverse geopolitical events

  2. act:  Actual adverse geopolitical events that have occurred or are actively
     occurring (wars, attacks, terrorist acts, invasions, coups, etc.)

  3. both:  Substantive discussion of both realized events AND associated
     threats/risks

Rules:
- Choose ONLY ONE: 'threat', 'act', or 'both'
- Be precise:  only choose 'both' if the article substantially discusses BOTH
  temporal dimensions
- Brief mentions of past context while focusing on future risks = 'threat'
- Brief mentions of future risks while focusing on current/past events = 'act'
```

### A.7.2 Oil GPR

The Oil GPR classification prompt first identifies whether an article talks about potential or realized oil supply disruptions. Relevant articles are then assigned regional labels based on where the oil supply disruption/threat is and which regions are affected. Multiple countries can be listed under the regional label.

```
OIL-GPR CLASSIFICATION PROMPT:
Analyze this newspaper article about geopolitical risk.
TASK 1:  Determine if this article discusses events that could disrupt oil or
energy supplies.  Oil/Energy disruption includes:
- Military conflicts in oil-producing regions
- Sanctions on oil-producing countries
- Attacks on oil infrastructure, pipelines, refineries, or shipping routes
- Political instability in major oil producers
- Blockades of oil shipping lanes (Strait of Hormuz, Suez Canal, etc.)
- Natural disasters affecting oil production
- Oil embargoes or export restrictions
```

```
- Terrorist attacks on energy infrastructure
Answer "yes" if the article discusses events that could affect oil/energy
production, transportation, or supply.  Answer "no" if the article discusses
geopolitical events unlikely to affect oil/energy markets.

TASK 2:  If "yes", identify which oil/energy producing region(s) are affected.
Select one or more from this list ONLY: Middle_East, Russia, USA, Venezuela,
North_Africa, West_Africa, Central_Asia, North_Sea, Canada, Mexico, Latin_America,
Southeast_Asia, China, Other
REGIONAL MAPPING RULES:
- Use 'Middle_East' for:  Saudi Arabia, Iran, Iraq, Kuwait, UAE, Qatar, Bahrain,
  Oman, Yemen, Israel, Syria, Lebanon
- Use 'Russia' for:  Russia, Soviet Union, USSR, Russian Federation
- Use 'USA' for:  United States, America, USA, U.S. (as oil producer, not just
  consumer)
- Use 'Venezuela' for:  Venezuela
- Use 'North_Africa' for:  Libya, Algeria, Egypt, Tunisia
- Use 'West_Africa' for:  Nigeria, Angola, Equatorial Guinea, Gabon, Congo
- Use 'Central_Asia' for:  Kazakhstan, Azerbaijan, Turkmenistan, Uzbekistan
- Use 'North_Sea' for:  Norway, UK (when referring to North Sea oil production)
- Use 'Canada' for:  Canada (oil sands, Alberta, etc.)
- Use 'Mexico' for:  Mexico
- Use 'Latin_America' for:  Other Latin American oil producers (Brazil, Colombia,
  Ecuador, Argentina)
- Use 'Southeast_Asia' for:  Indonesia, Malaysia, Brunei, Vietnam
- Use 'China' for:  China (as oil producer)
- Use 'Other' for:  Any other oil/energy producing region not listed above
IMPORTANT:
- Focus on regions WHERE the disruption is occurring, not where it's being
  discussed
- Include ALL regions mentioned that could experience oil/energy disruption
- If article mentions global impact or multiple regions, list all relevant ones
- If uncertain whether it affects oil/energy, err on the side of ''yes'' if the
  region is a major producer
```

### A.7.3 Country GPR (and Event)

The country classifier prompt identifies the main countries involved in a geopolitical incident as portrayed in a given article. Relevant countries are assigned one of three actor labels indicating whether they are an initiator, a respondent, or a spillover party. The prompt also assigns an event-type label for the article's central geopolitical conflict.

```
COUNTRY GPR:
Analyze this newspaper article about a geopolitical risk event.
```

```
TASK: Identify the key country-level actors in this geopolitical event and assign
each a role.  Roles are defined as follows:

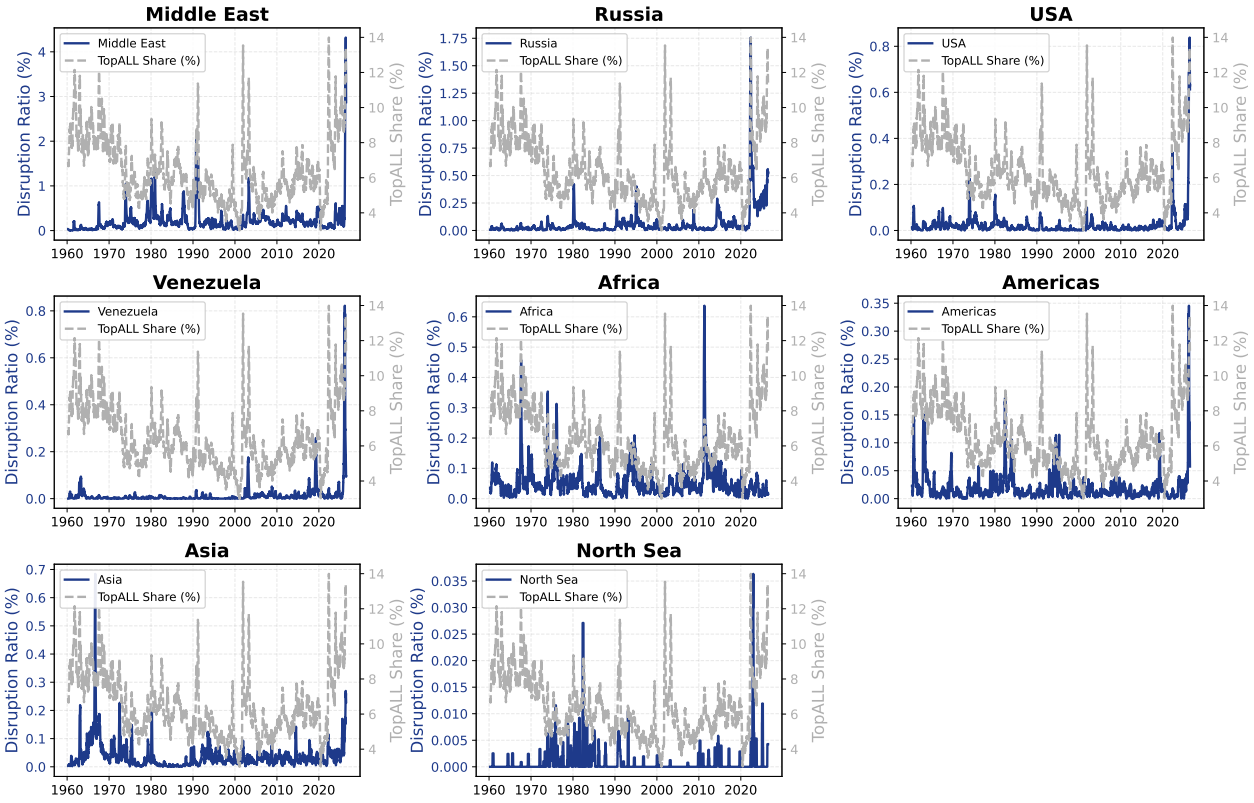
- initiator:  the country that launched or triggered the hostile action (military
  attack, invasion, sanctions, blockade, etc.)
- respondent:  the country that is the direct object of the hostile action
- spillover:  a country not directly involved but significantly affected (energy
  shock, refugee flows, trade disruption, etc.)

Rules:
- List up to 5 actors total; include only countries with a meaningful role in THIS
  article
- Use standard country names (e.g.  "United States", "Russia", "Saudi Arabia",
  "Israel")
- If a supra-national body (NATO, UN, EU) is the dominant actor, use its name
- Assign exactly one role per actor
- When initiation is genuinely contested or unclear (civil wars, mutual
  escalations, disputed territories), assign both main parties the role of
  initiator
- If the article is too vague to identify actors, return an empty actors list
Respond in JSON format only:
{"actors":  [
  {"country":  "<name>", "role":  "<role>"},
  ...
],
"event_type":  "<one of:  military_conflict, sanctions, terrorism,
diplomatic_tension, coup, civil_war, nuclear_threat, other>"}
```

## A.8 GPR-Driven Oil Disruptions by Region

This appendix shows a decomposition of the Oil GPR index of Section 5.2 by geographic region. For each region, we apply the same two-stage LLM classification used for the aggregate index but retain only articles in which the oil or energy disruption occurs in, or affects, that region; each series is the score-weighted count of such articles divided by total newspaper volume. Figure A.3 reports the eight regional panels.

Figure A.3: GPR-Driven Oil Disruptions by Region




*Notes*: Each panel shows the regional Oil GPR index (90-day moving average) for a specific geographic region. The index is computed as the ratio of score-weighted articles about oil disruptions in that region to total newspaper articles. The gray dashed line in each panel shows the overall AI-GPR share for reference. Regions are: Middle East, Russia, USA, Venezuela, Africa (North + West), Americas (Canada, Mexico, Latin America), Asia (Central, Southeast, China), and North Sea. The Middle East, which includes Iran, dominates the Oil GPR in the 1990s and again in 2026; Russia stands out after 2014, and Venezuela during its political crises. Source: authors' calculations. Last observation: June 2026.

## A.9 Data Details for the Oil Analysis

This appendix collects the data sources behind Figure 5 and the estimation choices behind the proxy VAR of Section 6.2. In the figure, the oil price is the WTI spot price, FRED series WTISPLC, topped up for the current partial month with the daily series DCOILWTICO, and deflated by the consumer price index, FRED series CPIAUCSL, to the dollars of the last month plotted. The system is estimated in levels with a constant and 12 lags, on monthly data from January 1985 to June 2026. The oil price is the U.S. crude oil first purchase price published by the Energy Information Administration, in nominal dollars per barrel and entered in logs. Global oil production is world crude oil including lease condensate, in thousand barrels per day, from the same source. Consumer prices enter in logs and the ten-year Treasury yield in levels, both from FRED. The daily oil price used to build the instrument is a different series: it is the WTI spot price at Cushing, which is quoted daily and is available from 1986 onward.

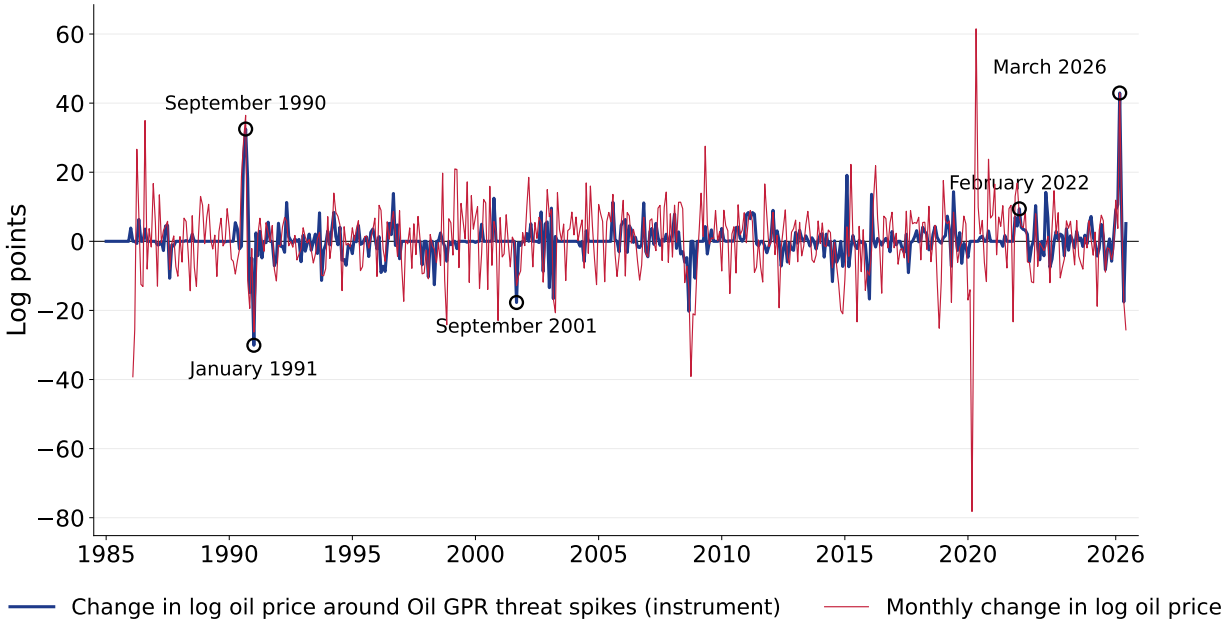
Monthly GDP is the Brave-Butters-Kelley index, which is published as an annualized monthly

log difference. We divide it by 1,200 to convert it to a monthly log change and cumulate it, so that it enters the VAR as a log level comparable to the other variables. The Oil GPR threats index enters the VAR as a monthly average of the daily series, standardized over the full history of the index.

The October 2025 consumer price index, not published because of the government shutdown, is interpolated linearly.

Flagged days are frequent enough that the instrument is not sparse: the Oil GPR threats index exceeds one standard deviation on about nine percent of days, which leaves roughly two thirds of the months in the sample with a non-zero instrument. Figure A.4 plots the instrument against the total monthly change in the oil price. In the largest geopolitical episodes, the 1990–91 Gulf crisis and the 2026 war with Iran, the instrument accounts for most of the monthly move, while large price swings with other causes, such as the collapses of 2014–15 and 2020, leave it close to zero.

Figure A.4: Oil Price Instrument and Monthly Oil Price Changes




*Notes*: The thin red line is the monthly change in the log WTI spot oil price. The thick blue line is the instrument, the part of that change that occurs around spikes in the Oil GPR threats index (Section 6.2). Both are in log points. Circles mark selected months. The instrument is zero before daily prices start in 1986. Source: authors' calculations; oil prices from FRED. Last observation: June 2026.

Confidence bands come from the moving block bootstrap of Jentsch and Lunsford (2022) with 6-month blocks and 500 draws.